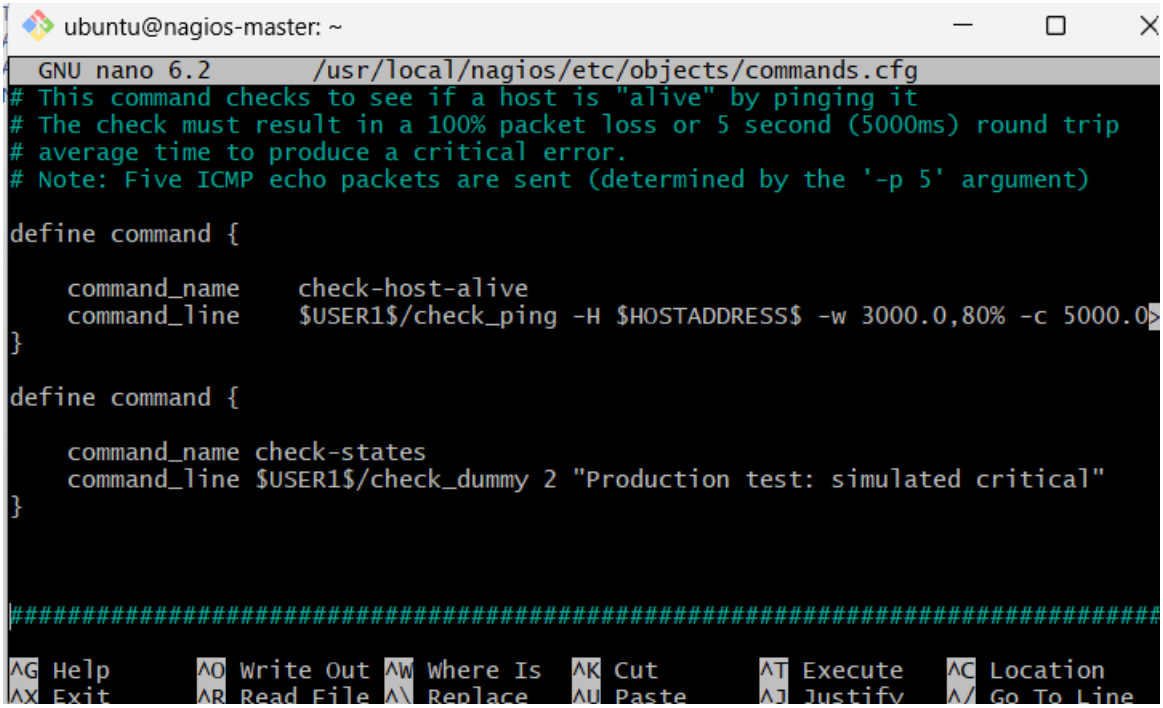
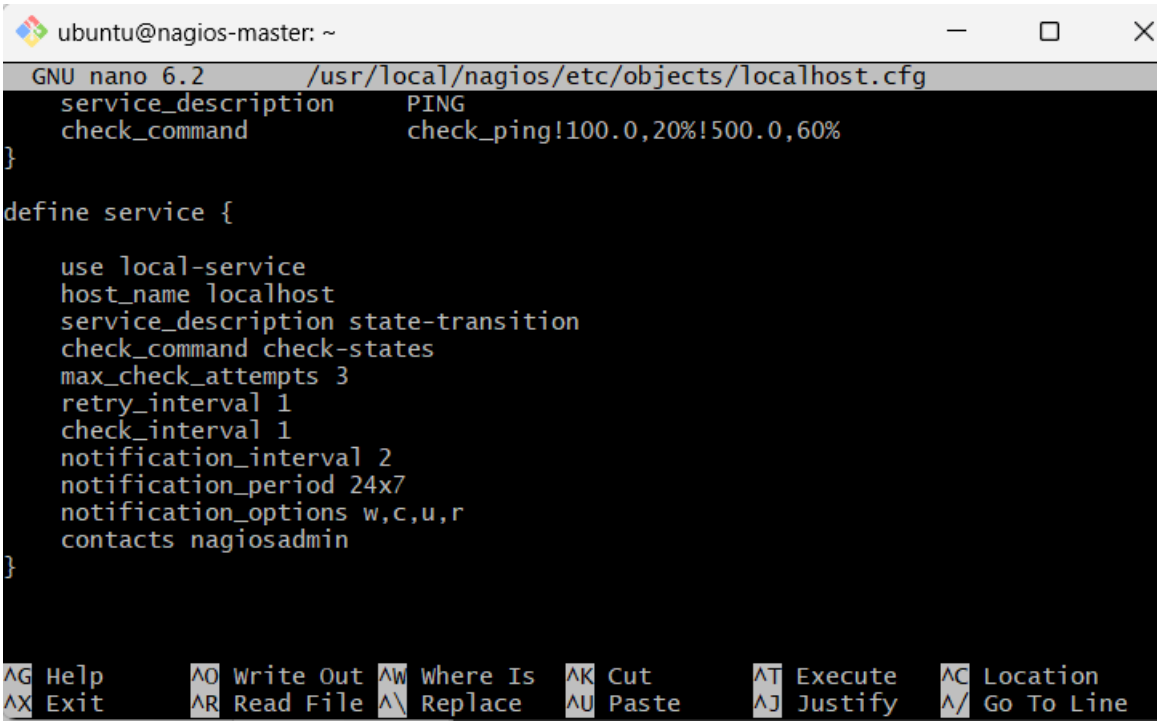
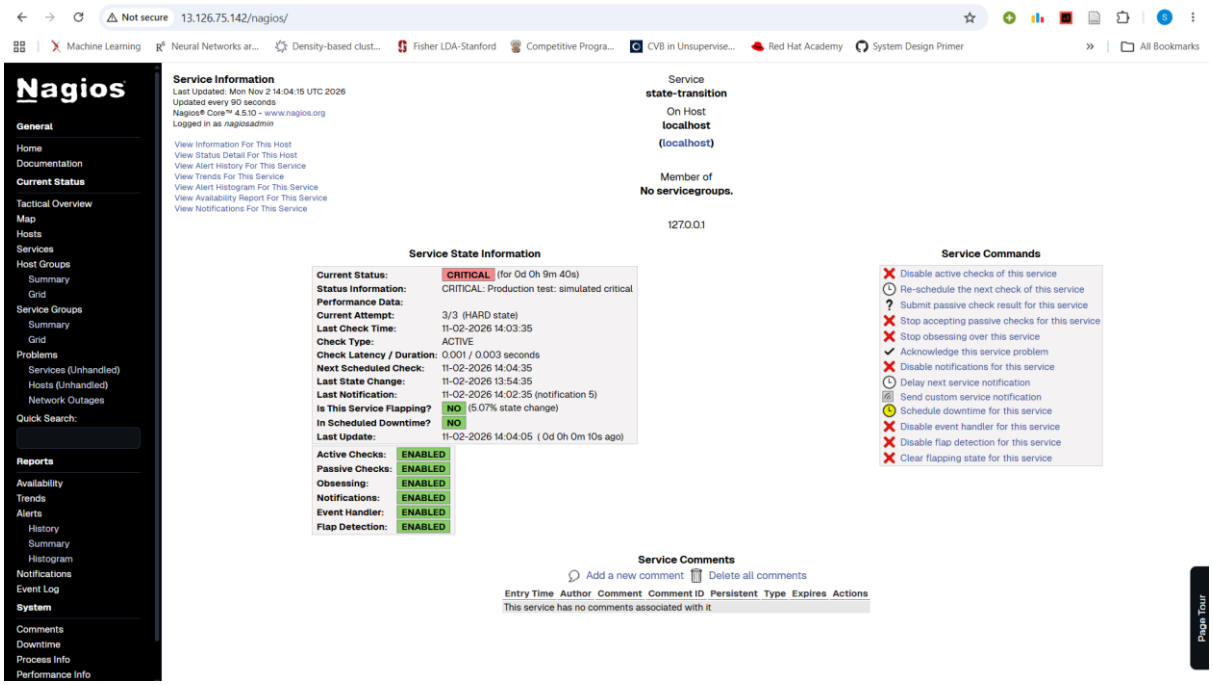
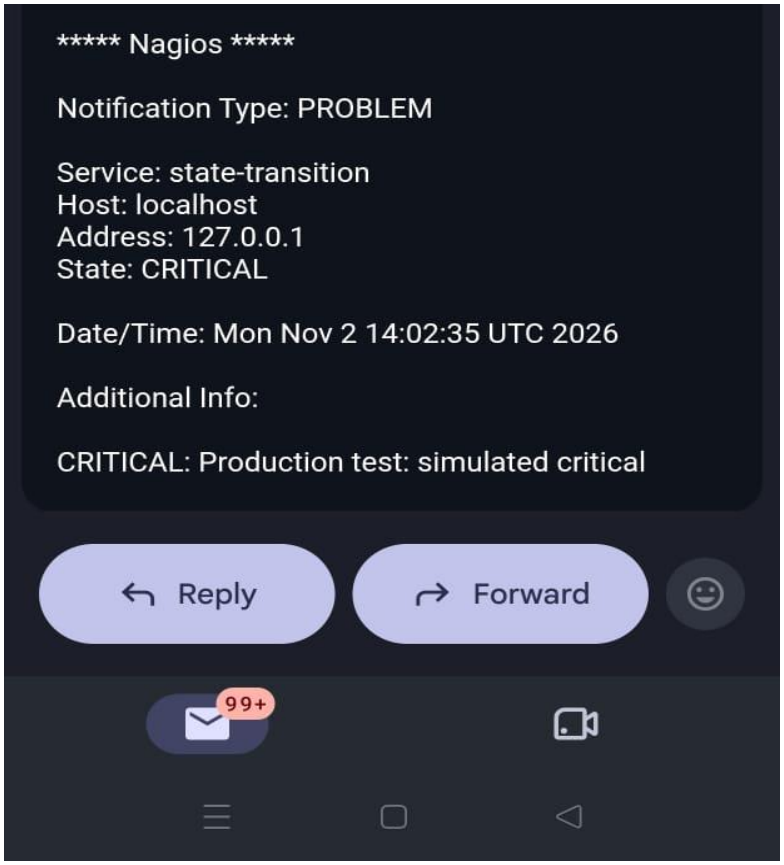
Nagios Final Armageddon Preparation (Extensive Practice)-Part 1(Includes the Puppet-Nagios practical integration from page 16)





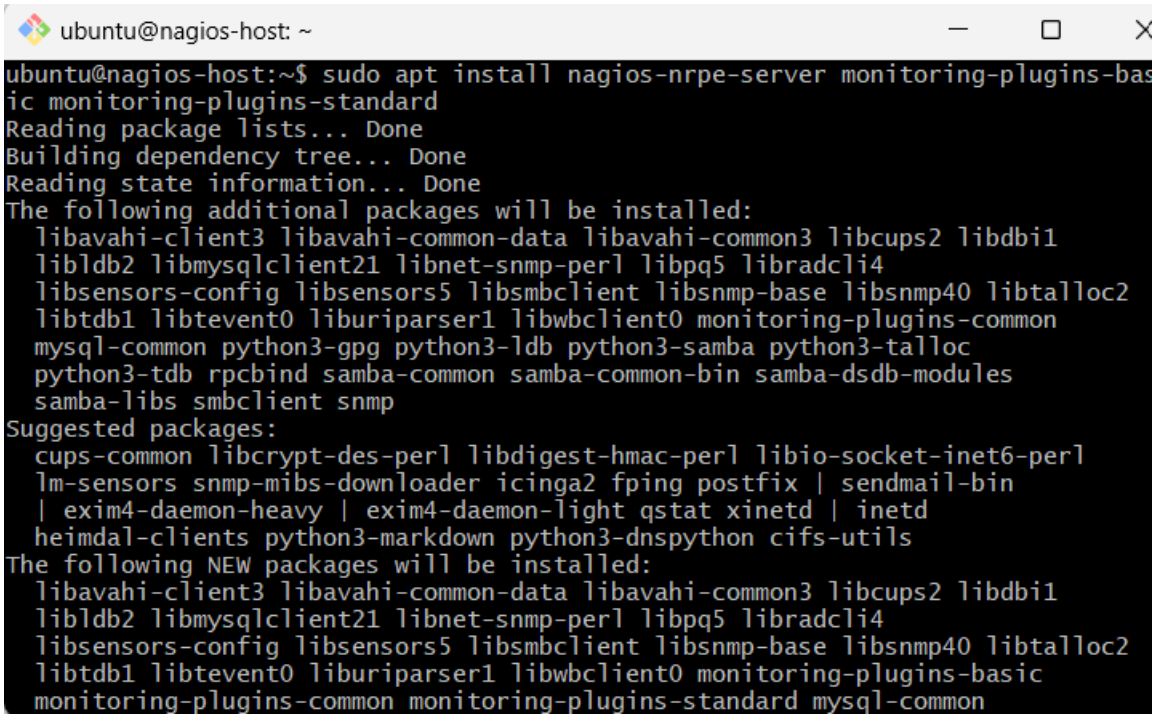
Let’s go for the criticality simulation:

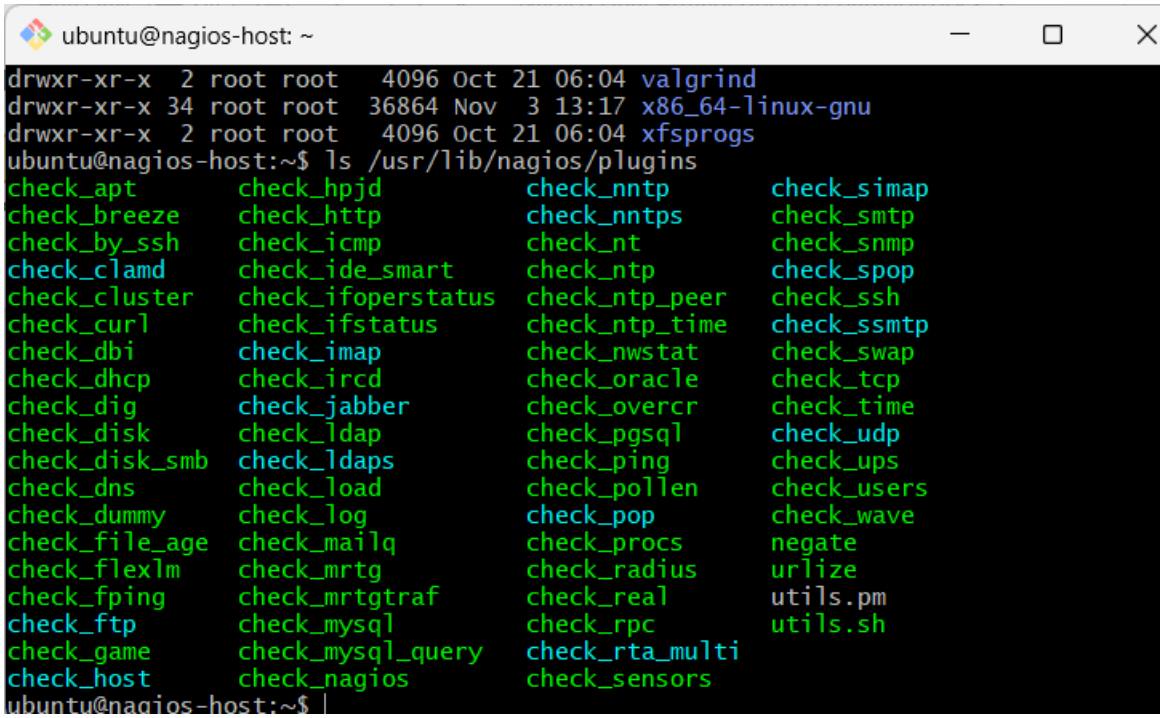


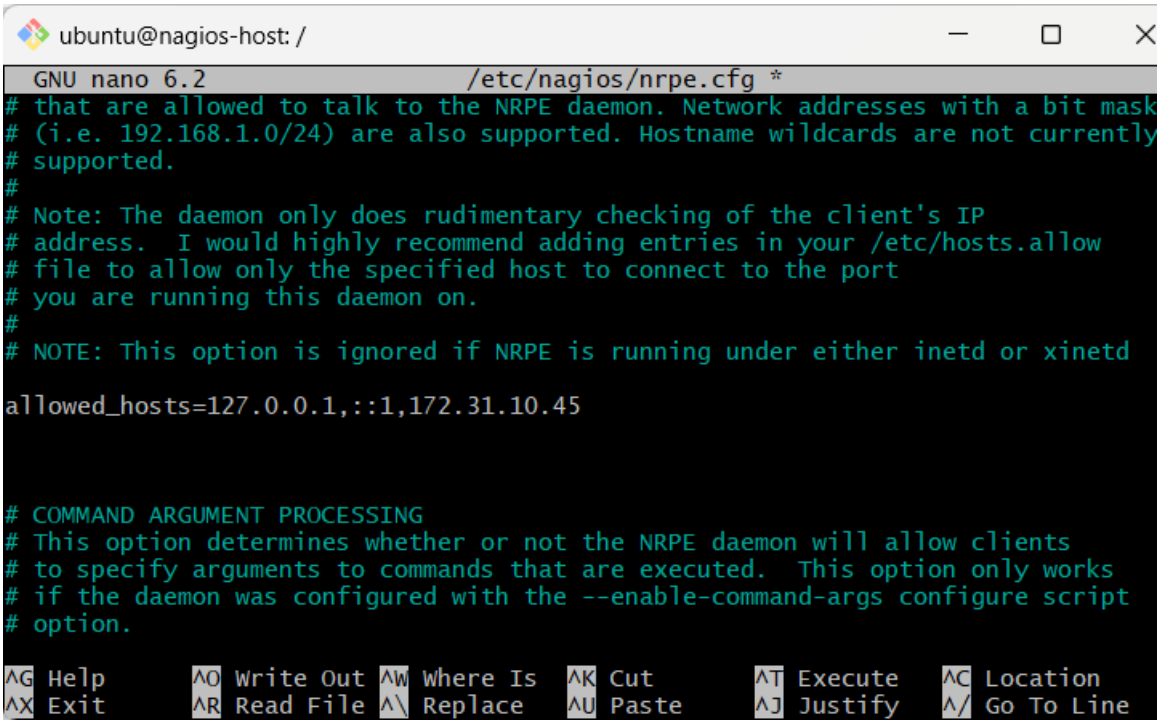


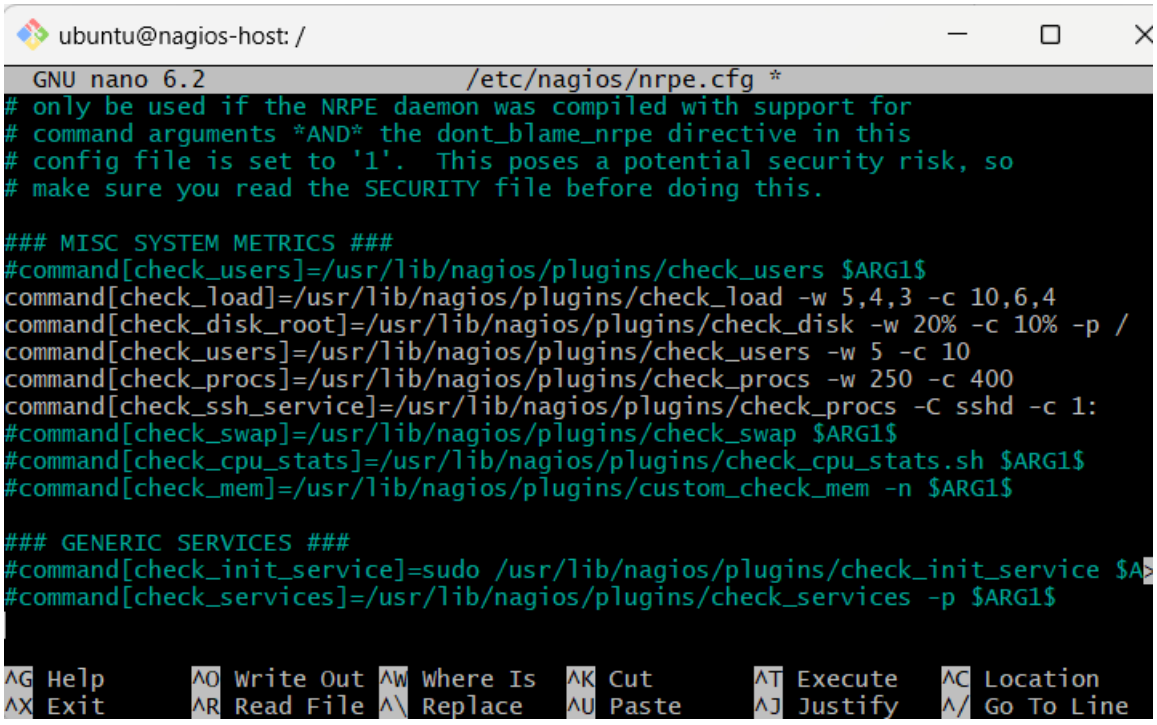
Nagios Practical simul via NRPE:

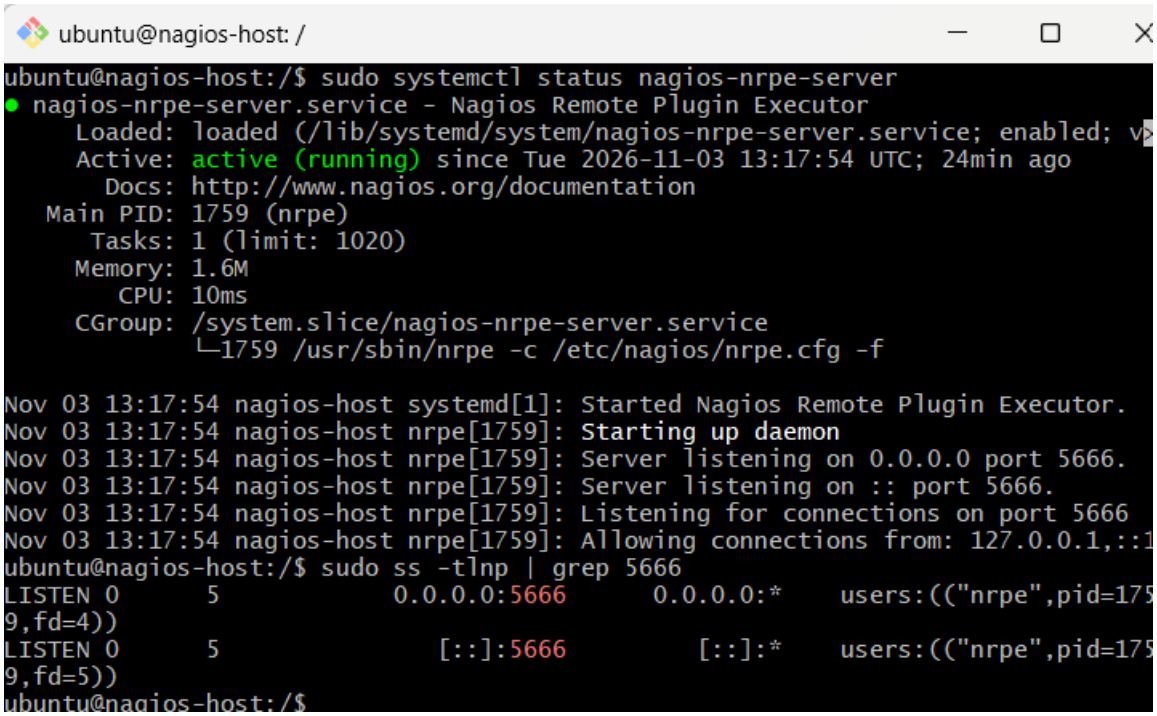
Recap of core steps for installation of NRPE:

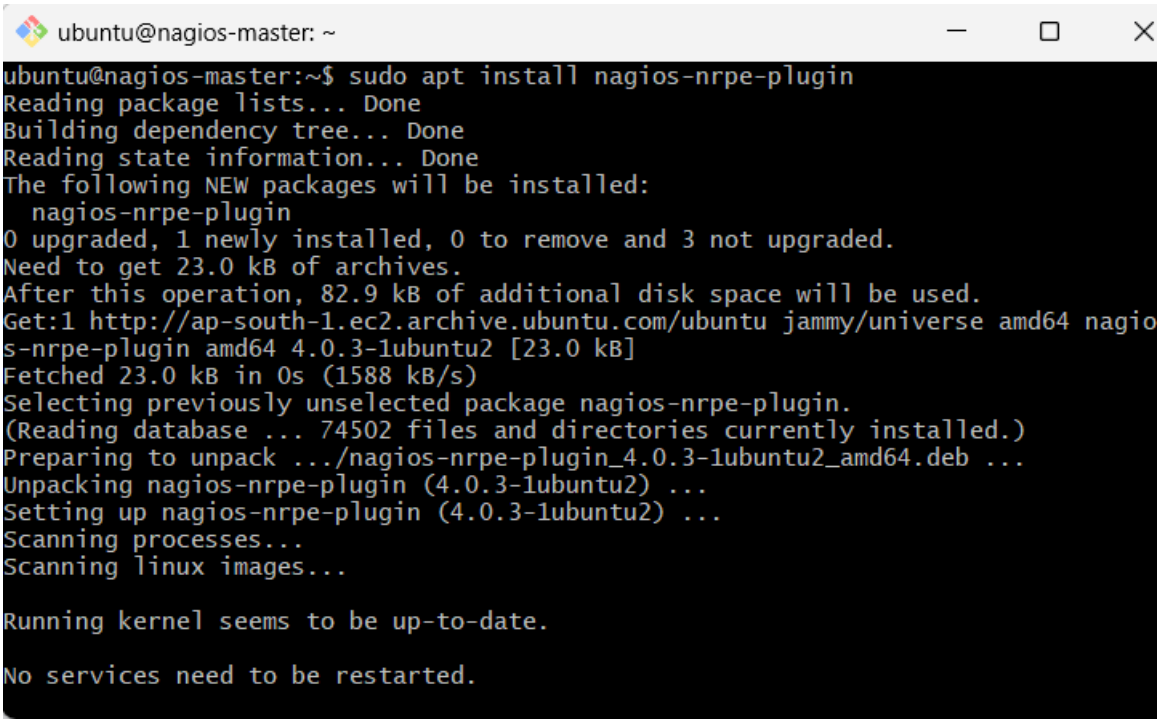

ubuntu@nagios-host: ~
ubuntu@nagios-host:~$ sudo apt install nagios-nrpe-server monitoring-plugins-bas
ic monitoring-plugins-standard
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
libavahi-client3 libavahi-common-data libavahi-common3 libcups2 libdbi1
libldb2 libmysqlclient21 libnet-snmp-perl libpq5 libradcli4
libsensors-config libsensors5 libsmbclient libsnmp-base libsnmp40 libtalloc2
libtdb1 libtevent0 liburiparser1 libwbclient0 monitoring-plugins-common
mysql-common python3-gpg python3-ldb python3-samba python3-talloc
python3-tdb rpcbind samba-common samba-common-bin samba-dsdb-modules
samba-libs smbclient snmp
Suggested packages:
cups-common libcrypt-des-perl libdigest-hmac-perl libio-socket-inet6-perl
lm-sensors snmp-mibs-downloader icinga2 fping postfix | sendmail-bin
| exim4-daemon-heavy | exim4-daemon-light qstat xinetd | inetd
heimdal-clients python3-markdown python3-dnspython cifs-utils
The following NEW packages will be installed:
libavahi-client3 libavahi-common-data libavahi-common3 libcups2 libdbi1
libldb2 libmysqlclient21 libnet-snmp-perl libpq5 libradcli4
libsensors-config libsensors5 libsmbclient libsnmp-base libsnmp40 libtalloc2
libtdb1 libtevent0 liburiparser1 libwbclient0 monitoring-plugins-basic
monitoring-plugins-common monitoring-plugins-standard mysql-common

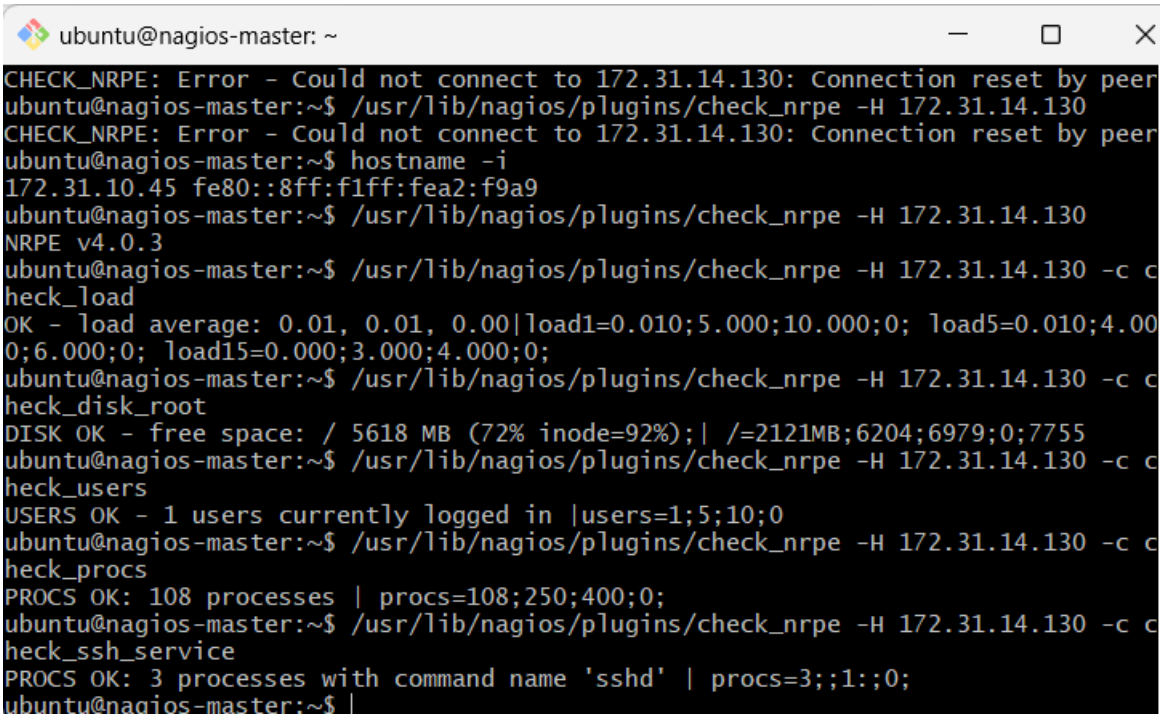

ubuntu@nagios-host: ~
drwxr-xr-x 2 root root 4096 Oct 21 06:04 valgrind
drwxr-xr-x 34 root root 36864 Nov 3 13:17 x86_64-linux-gnu
drwxr-xr-x 2 root root 4096 Oct 21 06:04 xfsprogs
ubuntu@nagios-host:~$ ls /usr/lib/nagios/plugins
check_apt check_hpjd check_nntp check_simap
check_breeze check_http check_nntps check_smtp
check_by_ssh check_icmp check_nt check_snmp
check_clamd check_ide_smart check_ntp check_spop
check_cluster check_ifoperstatus check_ntp_peer check_ssh
check_curl check_ifstatus check_ntp_time check_ssmtp
check_dbi check_imap check_nwstat check_swap
check_dhcp check_ircd check_oracle check_tcp
check_dig check_jabber check_overcr check_time
check_disk check_ldap check_pgsql check_udp
check_disk_smb check_ldaps check_ping check_ups
check_dns check_load check_pollen check_users
check_dummy check_log check_pop check_wave
check_file_age check_mailq check_procs negate
check_flexlm check_mrtg check_radius urlize
check_fping check_mrtgtraf check_real utils.pm
check_ftp check_mysql check_rpc utils.sh
check_game check_mysql_query check_rta_multi
check_host check_nagios check_sensors
ubuntu@nagios-host:~$

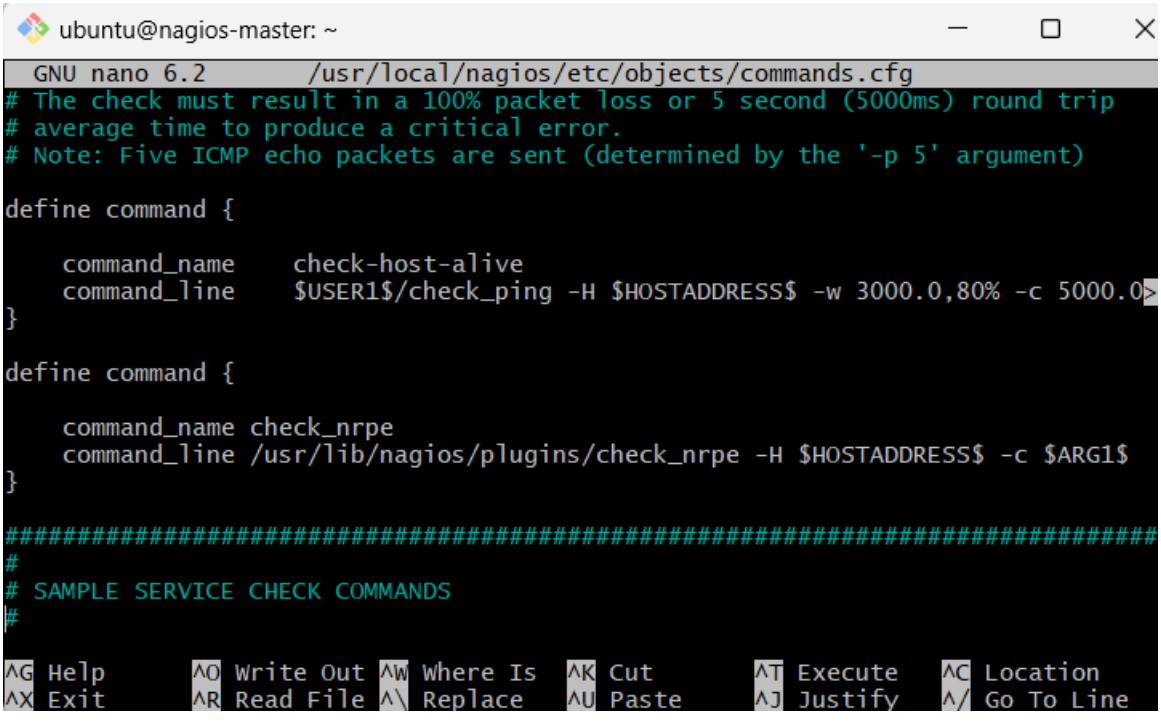

ubuntu@nagios-host: /
GNU nano 6.2 /etc/nagios/nrpe.cfg *
that are allowed to talk to the NRPE daemon. Network addresses with a bit mask
(i.e. 192.168.1.0/24) are also supported. Hostname wildcards are not currently
supported.
#
Note: The daemon only does rudimentary checking of the client's IP
address. I would highly recommend adding entries in your /etc/hosts.allow
file to allow only the specified host to connect to the port
you are running this daemon on.
#
NOTE: This option is ignored if NRPE is running under either inetd or xinetd
allowed_hosts=127.0.0.1,::1,172.31.10.45
COMMAND ARGUMENT PROCESSING
This option determines whether or not the NRPE daemon will allow clients
to specify arguments to commands that are executed. This option only works
if the daemon was configured with the --enable-command-args configure script
option.
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

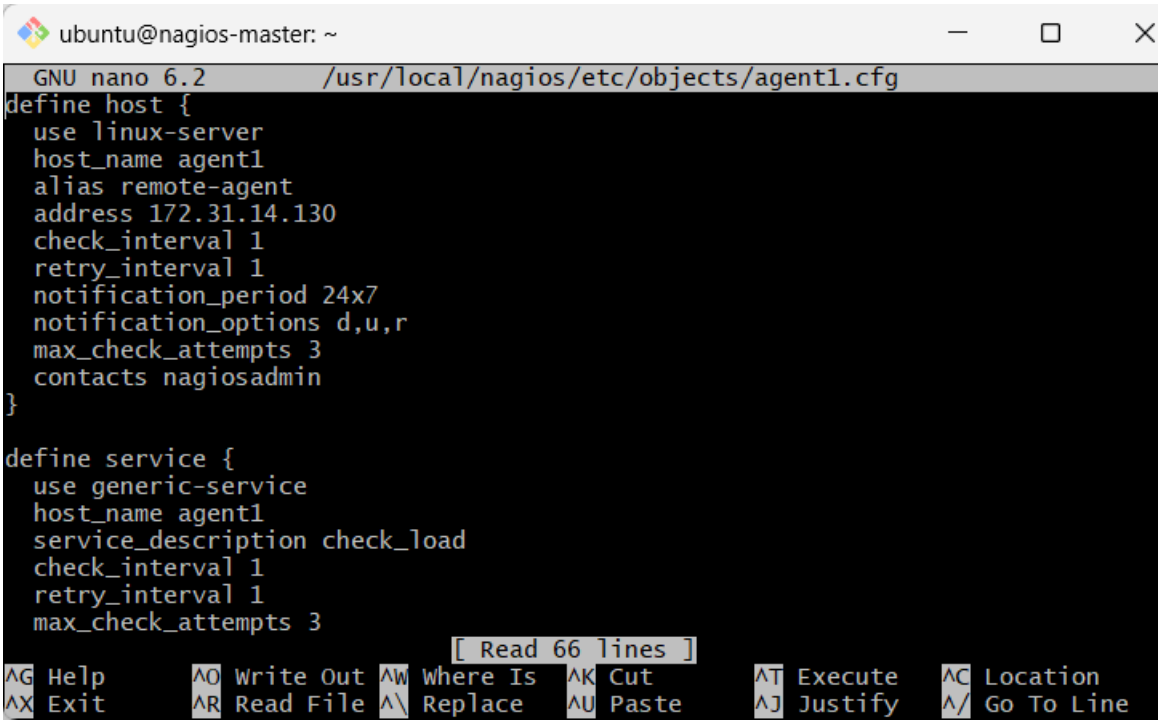

ubuntu@nagios-host: /
GNU nano 6.2 /etc/nagios/nrpe.cfg *
only be used if the NRPE daemon was compiled with support for
command arguments *AND* the dont_blame_nrpe directive in this
config file is set to '1'. This poses a potential security risk, so
make sure you read the SECURITY file before doing this.
MISC SYSTEM METRICS
#command[check_users]=/usr/lib/nagios/plugins/check_users $ARG1$
command[check_load]=/usr/lib/nagios/plugins/check_load -w 5,4,3 -c 10,6,4
command[check_disk_root]=/usr/lib/nagios/plugins/check_disk -w 20% -c 10% -p /
command[check_users]=/usr/lib/nagios/plugins/check_users -w 5 -c 10
command[check_procs]=/usr/lib/nagios/plugins/check_procs -w 250 -c 400
command[check_ssh_service]=/usr/lib/nagios/plugins/check_procs -C sshd -c 1:
#command[check_swap]=/usr/lib/nagios/plugins/check_swap $ARG1$
#command[check_cpu_stats]=/usr/lib/nagios/plugins/check_cpu_stats.sh $ARG1$
#command[check_mem]=/usr/lib/nagios/plugins/custom_check_mem -n $ARG1$
GENERIC SERVICES
#command[check_init_service]=sudo /usr/lib/nagios/plugins/check_init_service $A
#command[check_services]=/usr/lib/nagios/plugins/check_services -p $ARG1$
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

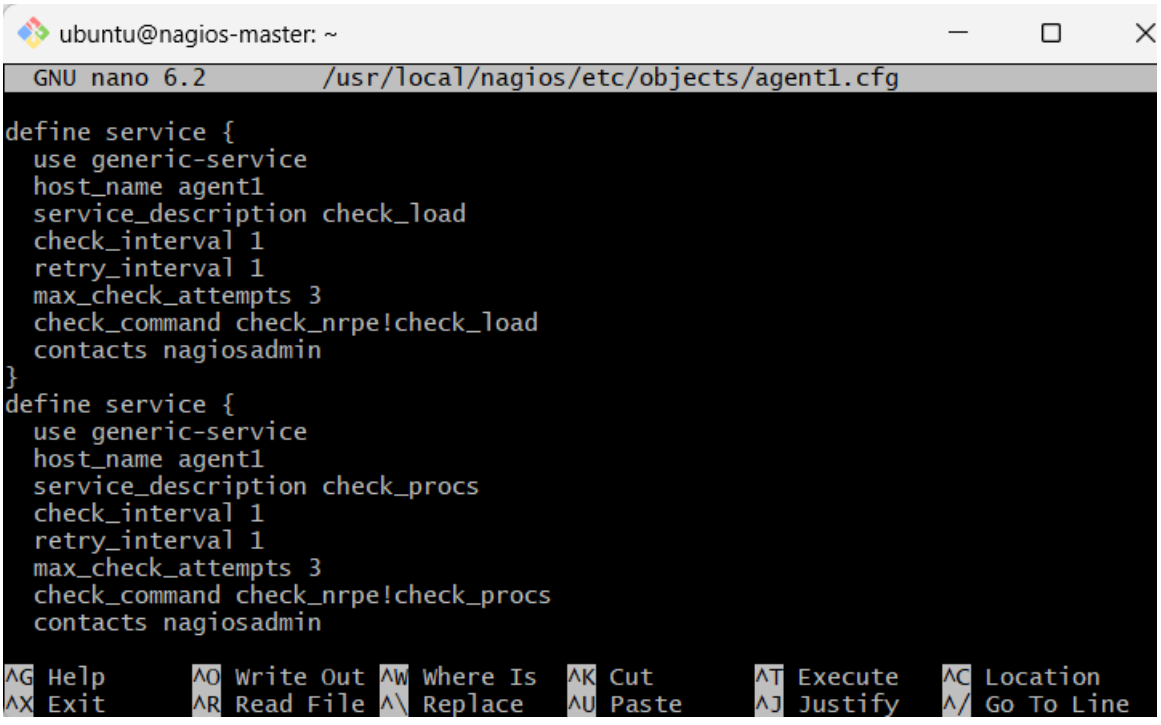

ubuntu@nagios-host: /
ubuntu@nagios-host:/$ sudo systemctl status nagios-nrpe-server
● nagios-nrpe-server.service - Nagios Remote Plugin Executor
Loaded: loaded (/lib/systemd/system/nagios-nrpe-server.service; enabled; v
Active: active (running) since Tue 2026-11-03 13:17:54 UTC; 24min ago
Docs: http://www.nagios.org/documentation
Main PID: 1759 (nrpe)
Tasks: 1 (limit: 1020)
Memory: 1.6M
CPU: 10ms
CGroup: /system.slice/nagios-nrpe-server.service
└─1759 /usr/sbin/nrpe -c /etc/nagios/nrpe.cfg -f
Nov 03 13:17:54 nagios-host systemd[1]: Started Nagios Remote Plugin Executor.
Nov 03 13:17:54 nagios-host nrpe[1759]: Starting up daemon
Nov 03 13:17:54 nagios-host nrpe[1759]: Server listening on 0.0.0.0 port 5666.
Nov 03 13:17:54 nagios-host nrpe[1759]: Server listening on :: port 5666.
Nov 03 13:17:54 nagios-host nrpe[1759]: Listening for connections on port 5666
Nov 03 13:17:54 nagios-host nrpe[1759]: Allowing connections from: 127.0.0.1,::1
ubuntu@nagios-host:/$ sudo ss -tlnp | grep 5666
LISTEN 0 5 0.0.0.0:5666 0.0.0.0:* users:(("nrpe",pid=175
9,fd=4))
LISTEN 0 5 [::]:5666 [::]:* users:(("nrpe",pid=175
9,fd=5))
ubuntu@nagios-host:/$

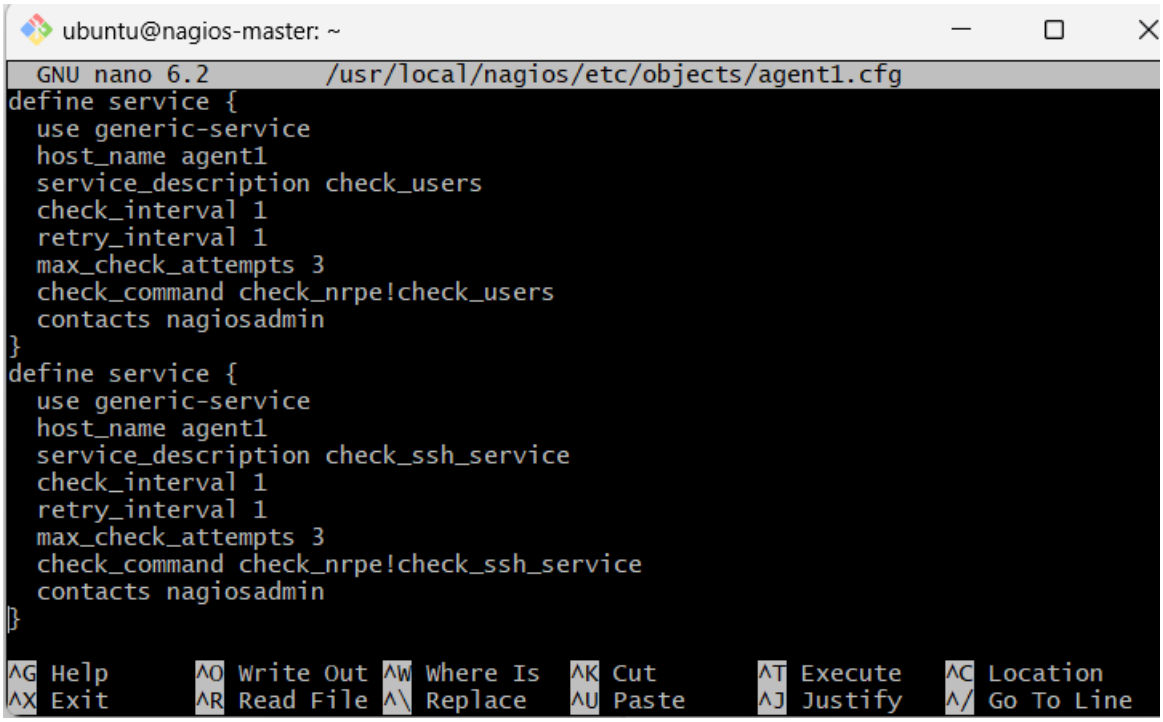

ubuntu@nagios-master: ~
ubuntu@nagios-master:~$ sudo apt install nagios-nrpe-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
nagios-nrpe-plugin
0 upgraded, 1 newly installed, 0 to remove and 3 not upgraded.
Need to get 23.0 kB of archives.
After this operation, 82.9 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 nagio
s-nrpe-plugin amd64 4.0.3-1ubuntu2 [23.0 kB]
Fetched 23.0 kB in 0s (1588 kB/s)
Selecting previously unselected package nagios-nrpe-plugin.
(Reading database ... 74502 files and directories currently installed.)
Preparing to unpack .../nagios-nrpe-plugin_4.0.3-1ubuntu2_amd64.deb ...
Unpacking nagios-nrpe-plugin (4.0.3-1ubuntu2) ...
Setting up nagios-nrpe-plugin (4.0.3-1ubuntu2) ...
Scanning processes...
Scanning linux images...
Running kernel seems to be up-to-date.
No services need to be restarted.

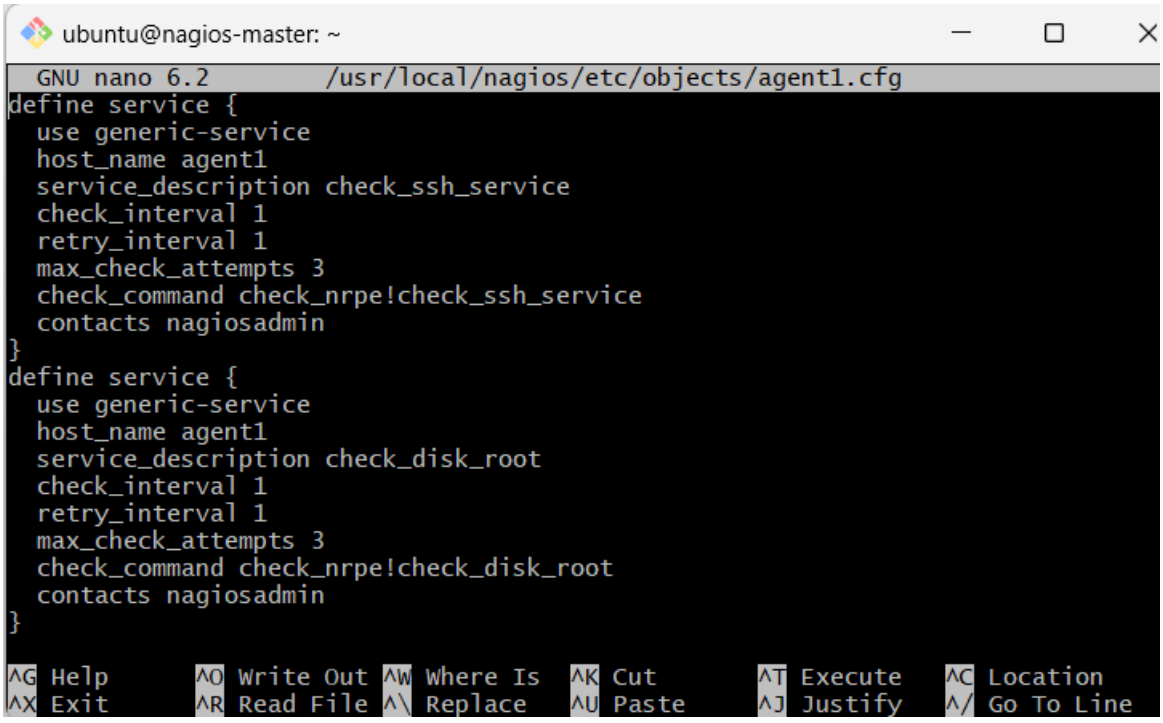

ubuntu@nagios-master: ~
CHECK_NRPE: Error - Could not connect to 172.31.14.130: Connection reset by peer
ubuntu@nagios-master:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.14.130
CHECK_NRPE: Error - Could not connect to 172.31.14.130: Connection reset by peer
ubuntu@nagios-master:~$ hostname -i
172.31.10.45 fe80::8ff:f1ff:fea2:f9a9
ubuntu@nagios-master:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.14.130
NRPE v4.0.3
ubuntu@nagios-master:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.14.130 -c c
heck_load
OK - load average: 0.01, 0.01, 0.00|load1=0.010;5.000;10.000;0; load5=0.010;4.00
0;6.000;0; load15=0.000;3.000;4.000;0;
ubuntu@nagios-master:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.14.130 -c c
heck_disk_root
DISK OK - free space: / 5618 MB (72% inode=92%);| /=2121MB;6204;6979;0;7755
ubuntu@nagios-master:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.14.130 -c c
heck_users
USERS OK - 1 users currently logged in |users=1;5;10;0
ubuntu@nagios-master:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.14.130 -c c
heck_procs
PROCS OK: 108 processes | procs=108;250;400;0;
ubuntu@nagios-master:~$ /usr/lib/nagios/plugins/check_nrpe -H 172.31.14.130 -c c
heck_ssh_service
PROCS OK: 3 processes with command name 'sshd' | procs=3;;1:;0;
ubuntu@nagios-master:~$

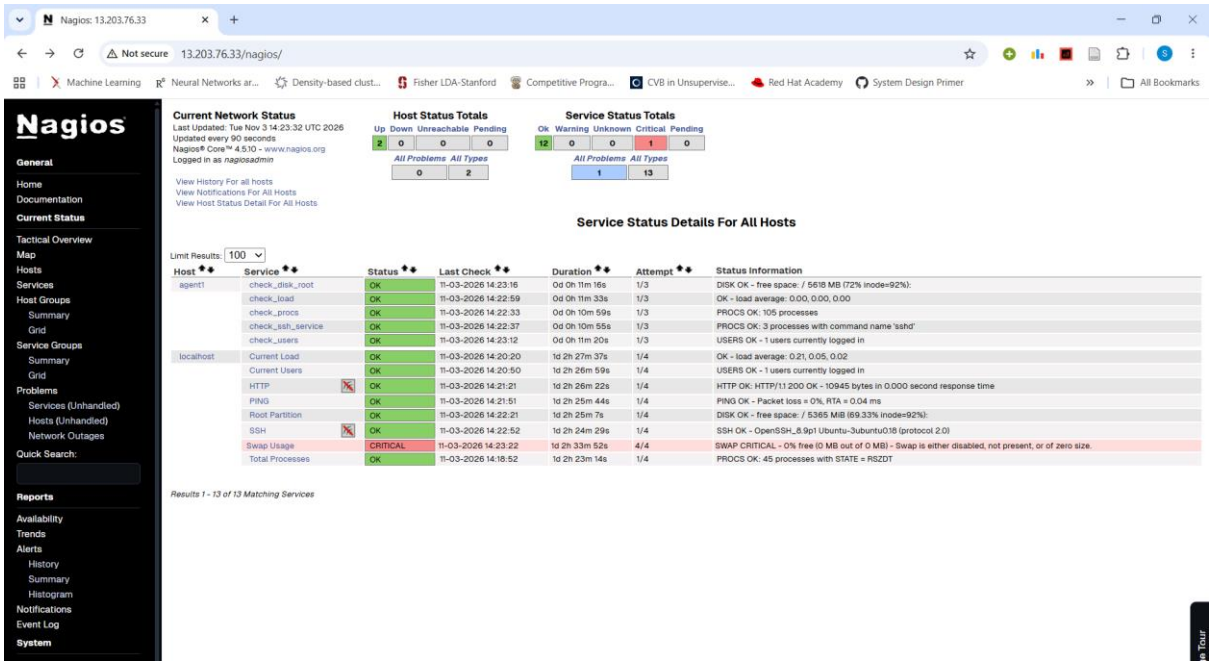

ubuntu@nagios-master: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/commands.cfg
The check must result in a 100% packet loss or 5 second (5000ms) round trip
average time to produce a critical error.
Note: Five ICMP echo packets are sent (determined by the '-p 5' argument)
define command {
command_name check-host-alive
command_line $USER1$/check_ping -H $HOSTADDRESS$ -w 3000.0,80% -c 5000.0
}
define command {
command_name check_nrpe
command_line /usr/lib/nagios/plugins/check_nrpe -H $HOSTADDRESS$ -c $ARG1$
}
#
SAMPLE SERVICE CHECK COMMANDS
#
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

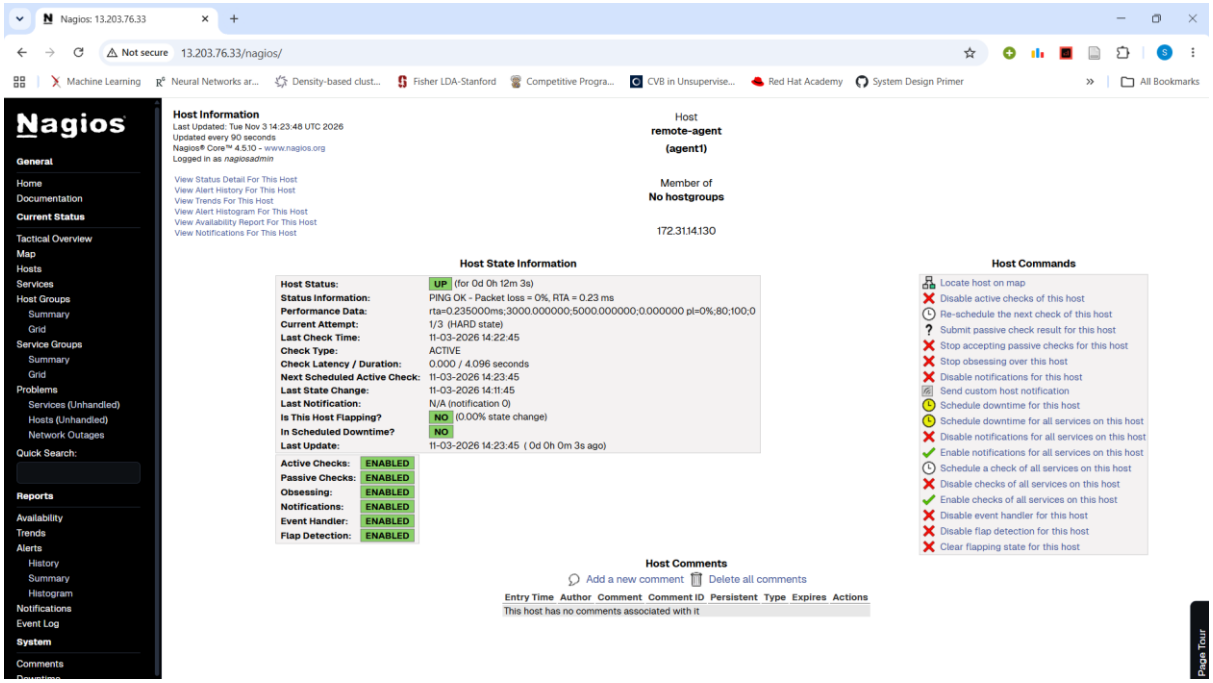

ubuntu@nagios-master: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/agent1.cfg
define host {
use linux-server
host_name agent1
alias remote-agent
address 172.31.14.130
check_interval 1
retry_interval 1
notification_period 24x7
notification_options d,u,r
max_check_attempts 3
contacts nagiosadmin
}
define service {
use generic-service
host_name agent1
service_description check_load
check_interval 1
retry_interval 1
max_check_attempts 3
[Read 66 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

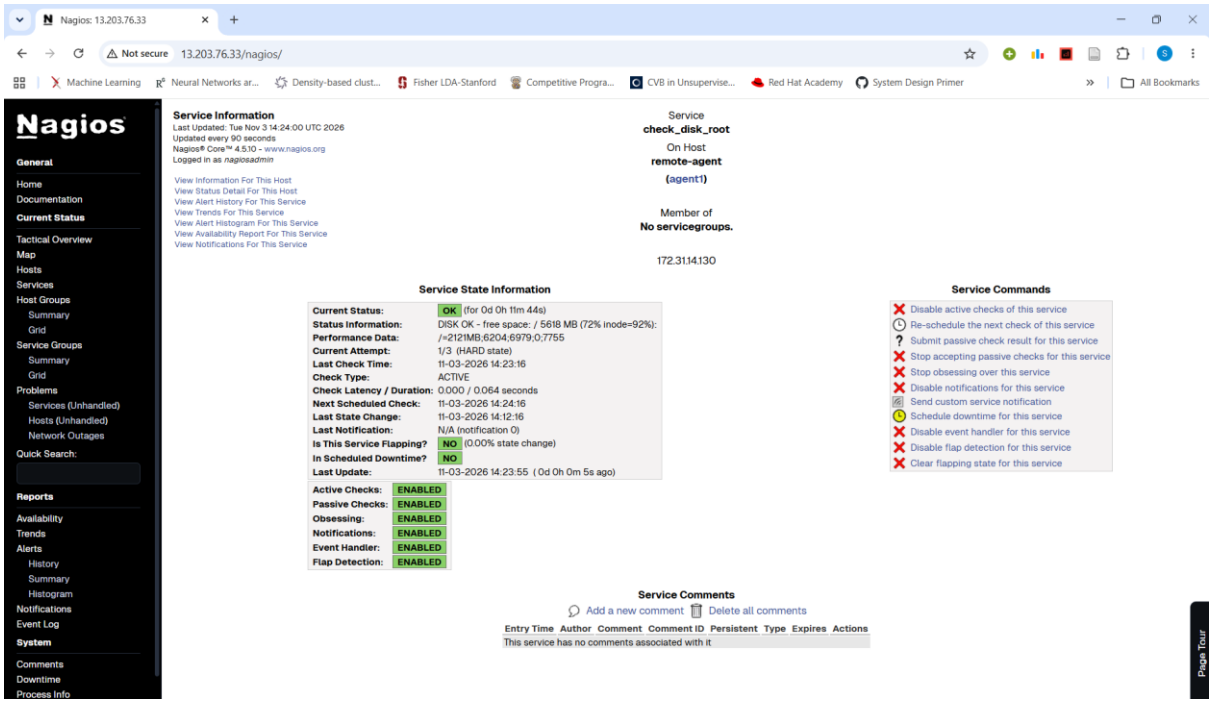

ubuntu@nagios-master: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/agent1.cfg
define service {
use generic-service
host_name agent1
service_description check_load
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_load
contacts nagiosadmin
}
define service {
use generic-service
host_name agent1
service_description check_procs
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_procs
contacts nagiosadmin
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

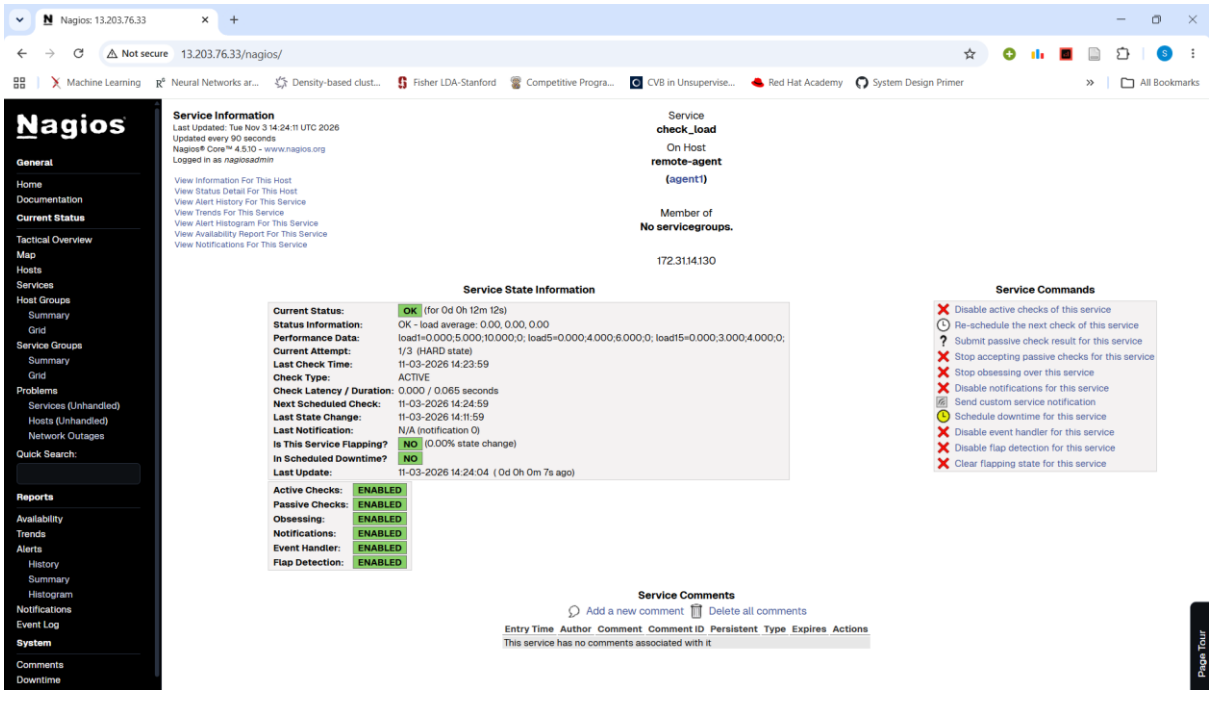

ubuntu@nagios-master: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/agent1.cfg
define service {
use generic-service
host_name agent1
service_description check_users
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_users
contacts nagiosadmin
}
define service {
use generic-service
host_name agent1
service_description check_ssh_service
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_ssh_service
contacts nagiosadmin
}
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

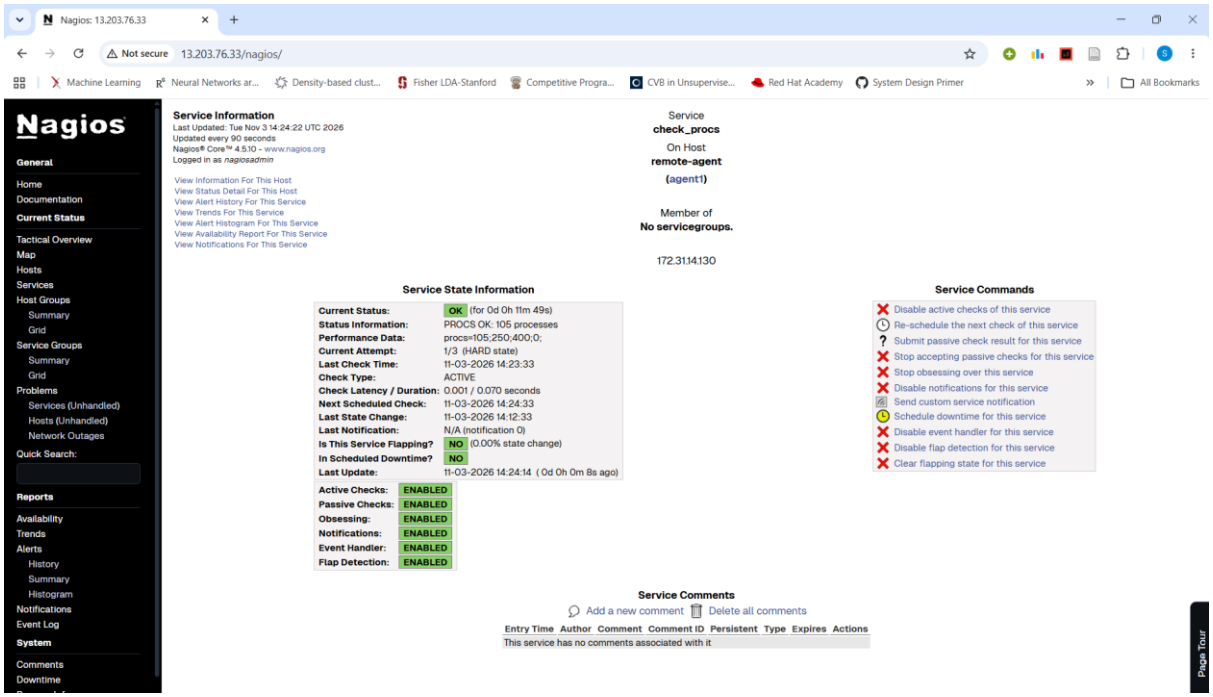

ubuntu@nagios-master: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/agent1.cfg
define service {
use generic-service
host_name agent1
service_description check_ssh_service
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_ssh_service
contacts nagiosadmin
}
define service {
use generic-service
host_name agent1
service_description check_disk_root
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_disk_root
contacts nagiosadmin
}
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

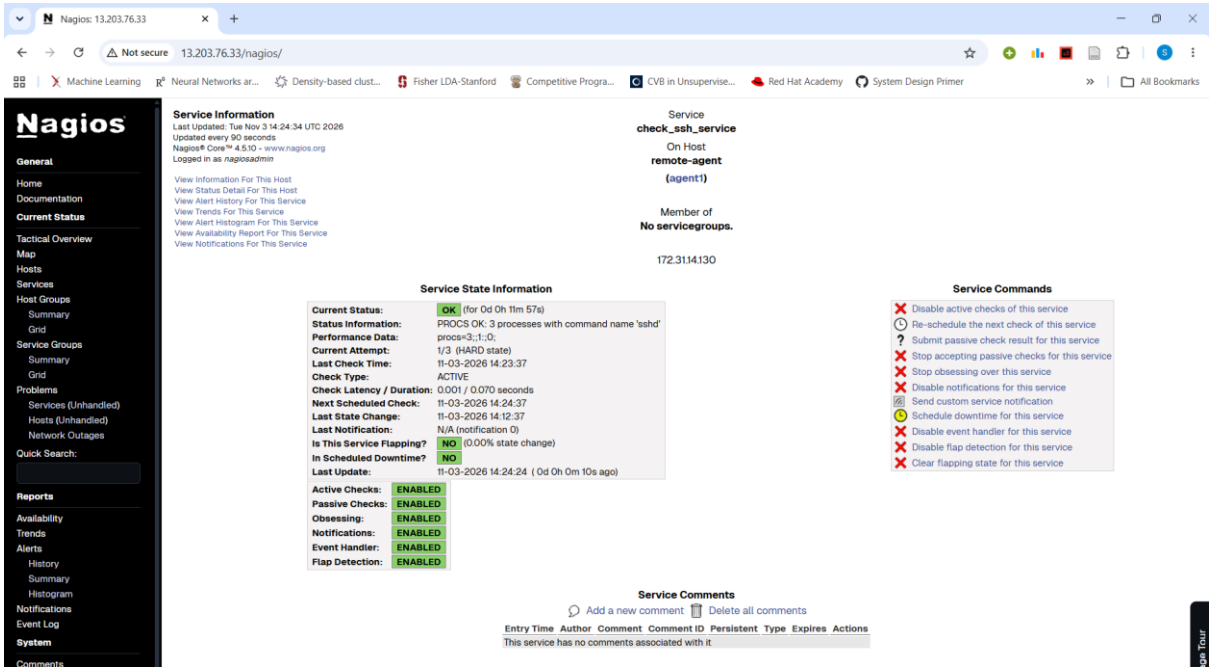

Nagios: 13.203.76.33
Not secure
13.203.76.33/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Current Network Status
Last Updated: Tue Nov 3 14:23:32 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin
View History For all hosts
View Notifications For All Hosts
View Host Status Detail For All Hosts
Host Status Totals
Up Down Unreachable Pending
2 0 0 0
All Problems All Types
0 2
Service Status Totals
Ok Warning Unknown Critical Pending
12 0 0 1 0
All Problems All Types
1 13
Service Status Details For All Hosts
Limit Results: 100
Host Service Status Last Check Duration Attempt Status Information
agent1 check_disk_root OK 11-03-2026 14:23:16 0d 0h 11m 16s 1/3 DISK OK - free space: / 5618 MB (72% inode=92%):
check_load OK 11-03-2026 14:22:59 0d 0h 11m 33s 1/3 OK - load average: 0.00, 0.00, 0.00
check_procs OK 11-03-2026 14:22:33 0d 0h 10m 59s 1/3 PROCS OK: 105 processes
check_ssh_service OK 11-03-2026 14:22:37 0d 0h 10m 55s 1/3 PROCS OK: 3 processes with command name 'sshd'
check_users OK 11-03-2026 14:23:12 0d 0h 11m 20s 1/3 USERS OK - 1 users currently logged in
localhost Current Load OK 11-03-2026 14:20:20 1d 2h 27m 37s 1/4 OK - load average: 0.21, 0.05, 0.02
Current Users OK 11-03-2026 14:20:50 1d 2h 26m 59s 1/4 USERS OK - 1 users currently logged in
HTTP OK 11-03-2026 14:21:21 1d 2h 26m 22s 1/4 HTTP OK: HTTP/1.1 200 OK - 10945 bytes in 0.000 second response time
PING OK 11-03-2026 14:21:51 1d 2h 25m 44s 1/4 PING OK - Packet loss = 0%, RTA = 0.04 ms
Root Partition OK 11-03-2026 14:22:21 1d 2h 25m 7s 1/4 DISK OK - free space: / 5365 MiB (69.33% inode=92%):
SSH OK 11-03-2026 14:22:52 1d 2h 24m 29s 1/4 SSH OK - OpenSSH_8.9p1 Ubuntu-3ubuntu0.18 (protocol 2.0)
Swap Usage CRITICAL 11-03-2026 14:23:22 1d 2h 33m 52s 4/4 SWAP CRITICAL - 0% free (0 MB out of 0 MB) - Swap is either disabled, not present, or of zero size.
Total Processes OK 11-03-2026 14:18:52 1d 2h 23m 14s 1/4 PROCS OK: 45 processes with STATE = RSZDT
Results 1 - 13 of 13 Matching Services

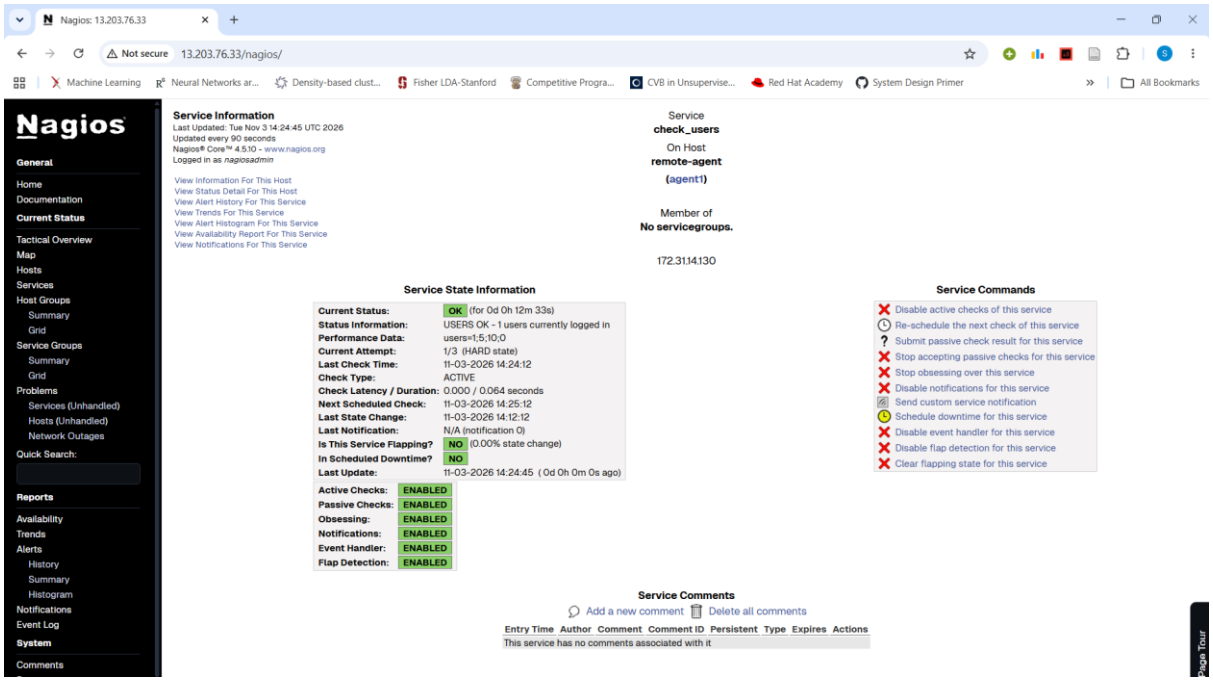

Nagios: 13.203.76.33
Not secure
13.203.76.33/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Comments
Host Information
Last Updated: Tue Nov 3 14:23:48 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin
View Status Detail For This Host
View Alert History For This Host
View Trends For This Host
View Alert Histogram For This Host
View Availability Report For This Host
View Notifications For This Host
Host
remote-agent
(agent1)
Member of
No hostgroups
172.31.14.130
Host State Information
Host Status: UP (for 0d 0h 12m 3s)
Status Information: PING OK - Packet loss = 0%, RTA = 0.23 ms
Performance Data: rta=0.235000ms;3000.000000;5000.000000;0.000000 pl=0%;80;100;0
Current Attempt: 1/3 (HARD state)
Last Check Time: 11-03-2026 14:22:45
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 4.096 seconds
Next Scheduled Active Check: 11-03-2026 14:23:45
Last State Change: 11-03-2026 14:11:45
Last Notification: N/A (notification 0)
Is This Host Flapping? NO (0.00% state change)
In Scheduled Downtime? NO
Last Update: 11-03-2026 14:23:45 (0d 0h 0m 3s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED
Host Commands
Locate host on map
Disable active checks of this host
Re-schedule the next check of this host
Submit passive check result for this host
Stop accepting passive checks for this host
Stop obsessing over this host
Disable notifications for this host
Send custom host notification
Schedule downtime for this host
Schedule downtime for all services on this host
Disable notifications for all services on this host
Enable notifications for all services on this host
Schedule a check of all services on this host
Disable checks of all services on this host
Enable checks of all services on this host
Disable event handler for this host
Disable flap detection for this host
Clear flapping state for this host
Host Comments
Add a new comment
Delete all comments
Entry Time Author Comment Comment ID Persistent Type Expires Actions
This host has no comments associated with it


Nagios: 13.203.76.33
Not secure
13.203.76.33/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Comments
Downtime
Process Info
Service Information
Last Updated: Tue Nov 3 14:24:00 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin
View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service
Service
check_disk_root
On Host
remote-agent
(agent1)
Member of
No servicegroups.
172.31.14.130
Service State Information
Current Status: OK (for 0d 0h 11m 44s)
Status Information: DISK OK - free space: / 5618 MB (72% inode=92%):
Performance Data: /=2121MB;6204;6979;0;7755
Current Attempt: 1/3 (HARD state)
Last Check Time: 11-03-2026 14:23:16
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 0.064 seconds
Next Scheduled Check: 11-03-2026 14:24:16
Last State Change: 11-03-2026 14:12:16
Last Notification: N/A (notification 0)
Is This Service Flapping? NO (0.00% state change)
In Scheduled Downtime? NO
Last Update: 11-03-2026 14:23:55 (0d 0h 0m 5s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED
Service Commands
Disable active checks of this service
Re-schedule the next check of this service
Submit passive check result for this service
Stop accepting passive checks for this service
Stop obsessing over this service
Disable notifications for this service
Send custom service notification
Schedule downtime for this service
Disable event handler for this service
Disable flap detection for this service
Clear flapping state for this service
Service Comments
Add a new comment
Delete all comments
Entry Time Author Comment Comment ID Persistent Type Expires Actions
This service has no comments associated with it
Page Tour

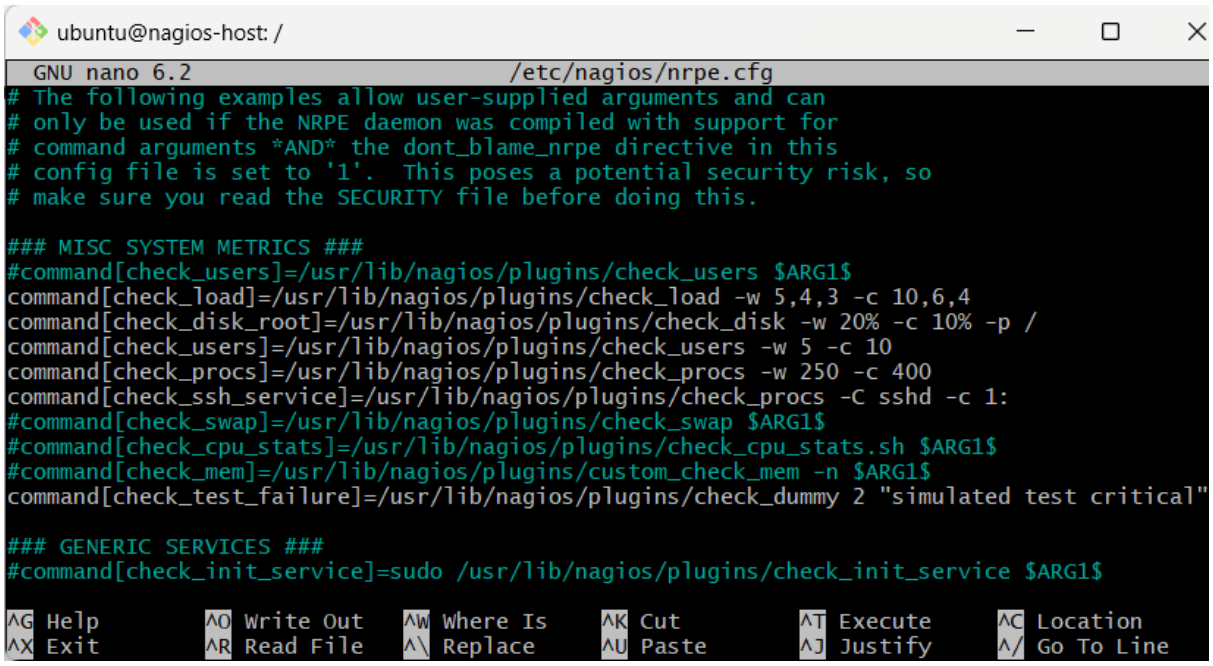

Nagios: 13.203.76.33
Not secure
13.203.76.33/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Comments
Downtime
Service Information
Last Updated: Tue Nov 3 14:24:11 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin
View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service
Service
check_load
On Host
remote-agent
(agent1)
Member of
No servicegroups.
172.31.14.130
Service State Information
Current Status: OK (for 0d 0h 12m 12s)
Status Information: OK - load average: 0.00, 0.00, 0.00
Performance Data: load1=0.000;5.000;10.000;0; load5=0.000;4.000;6.000;0; load15=0.000;3.000;4.000;0;
Current Attempt: 1/3 (HARD state)
Last Check Time: 11-03-2026 14:23:59
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 0.065 seconds
Next Scheduled Check: 11-03-2026 14:24:59
Last State Change: 11-03-2026 14:11:59
Last Notification: N/A (notification 0)
Is This Service Flapping? NO (0.00% state change)
In Scheduled Downtime? NO
Last Update: 11-03-2026 14:24:04 (0d 0h 0m 7s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED
Service Commands
Disable active checks of this service
Re-schedule the next check of this service
Submit passive check result for this service
Stop accepting passive checks for this service
Stop obsessing over this service
Disable notifications for this service
Send custom service notification
Schedule downtime for this service
Disable event handler for this service
Disable flap detection for this service
Clear flapping state for this service
Service Comments
Add a new comment
Delete all comments
Entry Time Author Comment Comment ID Persistent Type Expires Actions
This service has no comments associated with it
Page Tour

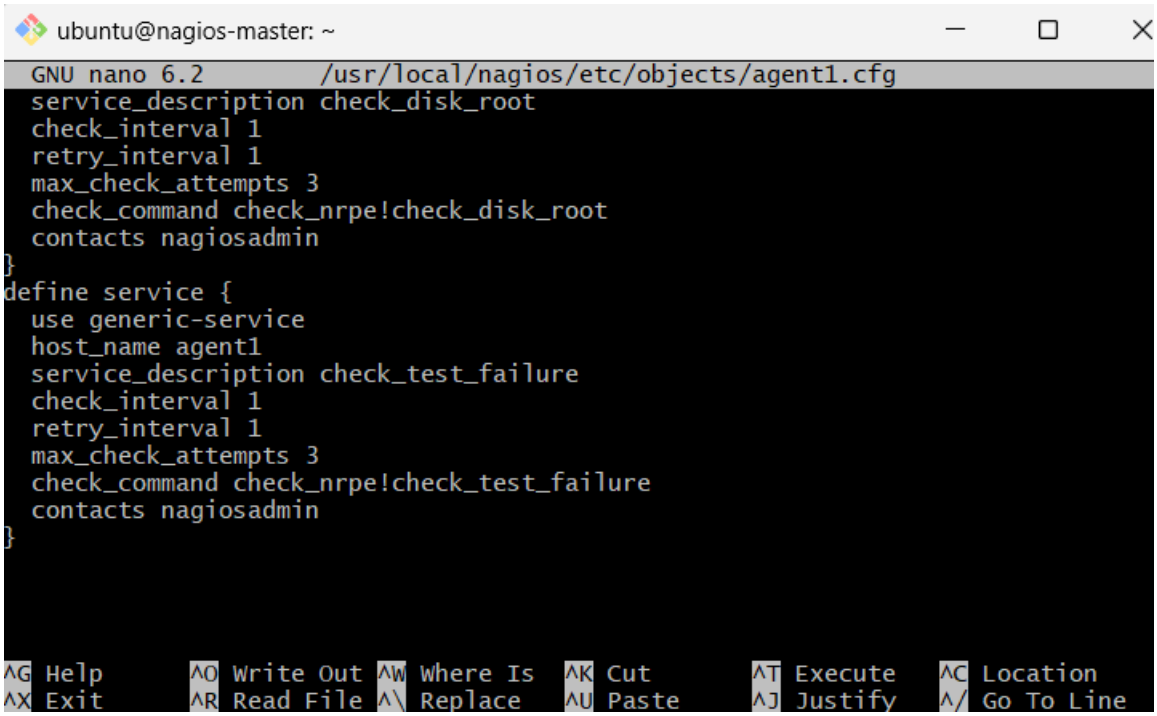

Nagios: 13.203.76.33
Not secure
13.203.76.33/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Comments
Downtime
Service Information
Last Updated: Tue Nov 3 14:24:22 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin
View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service
Service
check_procs
On Host
remote-agent
(agent1)
Member of
No servicegroups.
172.31.14.130
Service State Information
Current Status: OK (for 0d 0h 11m 49s)
Status Information: PROCS OK: 105 processes
Performance Data: procs=105;250;400;0;
Current Attempt: 1/3 (HARD state)
Last Check Time: 11-03-2026 14:23:33
Check Type: ACTIVE
Check Latency / Duration: 0.001 / 0.070 seconds
Next Scheduled Check: 11-03-2026 14:24:33
Last State Change: 11-03-2026 14:12:33
Last Notification: N/A (notification 0)
Is This Service Flapping? NO (0.00% state change)
In Scheduled Downtime? NO
Last Update: 11-03-2026 14:24:14 (0d 0h 0m 8s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED
Service Commands
Disable active checks of this service
Re-schedule the next check of this service
Submit passive check result for this service
Stop accepting passive checks for this service
Stop obsessing over this service
Disable notifications for this service
Send custom service notification
Schedule downtime for this service
Disable event handler for this service
Disable flap detection for this service
Clear flapping state for this service
Service Comments
Add a new comment
Delete all comments
Entry Time Author Comment Comment ID Persistent Type Expires Actions
This service has no comments associated with it
Page Tour

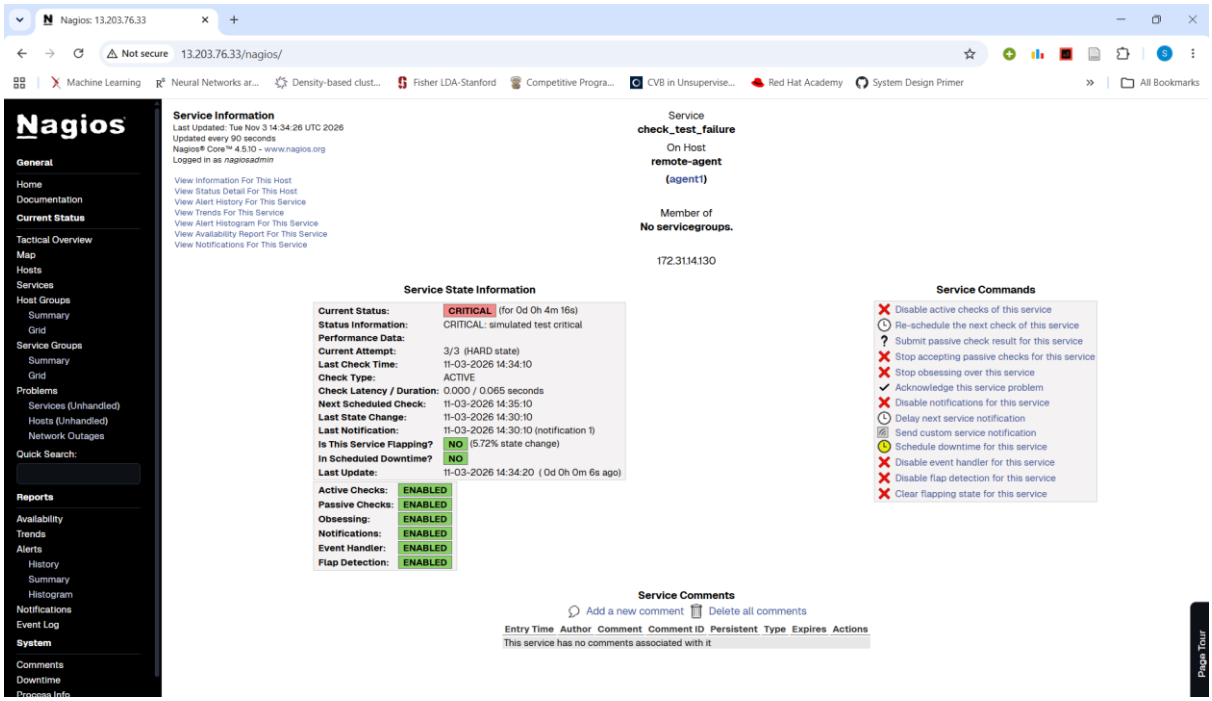

Nagios: 13.203.76.33
Not secure
13.203.76.33/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Comments
Service Information
Last Updated: Tue Nov 3 14:24:34 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin
View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service
Service
check_ssh_service
On Host
remote-agent
(agent1)
Member of
No servicegroups.
172.31.14.130
Service State Information
Current Status: OK (for 0d 0h 11m 57s)
Status Information: PROCS OK: 3 processes with command name 'sshd'
Performance Data: procs=3;;1;;0;
Current Attempt: 1/3 (HARD state)
Last Check Time: 11-03-2026 14:23:37
Check Type: ACTIVE
Check Latency / Duration: 0.001 / 0.070 seconds
Next Scheduled Check: 11-03-2026 14:24:37
Last State Change: 11-03-2026 14:12:37
Last Notification: N/A (notification 0)
Is This Service Flapping? NO (0.00% state change)
In Scheduled Downtime? NO
Last Update: 11-03-2026 14:24:24 (0d 0h 0m 10s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED
Service Commands
Disable active checks of this service
Re-schedule the next check of this service
Submit passive check result for this service
Stop accepting passive checks for this service
Stop obsessing over this service
Disable notifications for this service
Send custom service notification
Schedule downtime for this service
Disable event handler for this service
Disable flap detection for this service
Clear flapping state for this service
Service Comments
Add a new comment
Delete all comments
Entry Time Author Comment Comment ID Persistent Type Expires Actions
This service has no comments associated with it

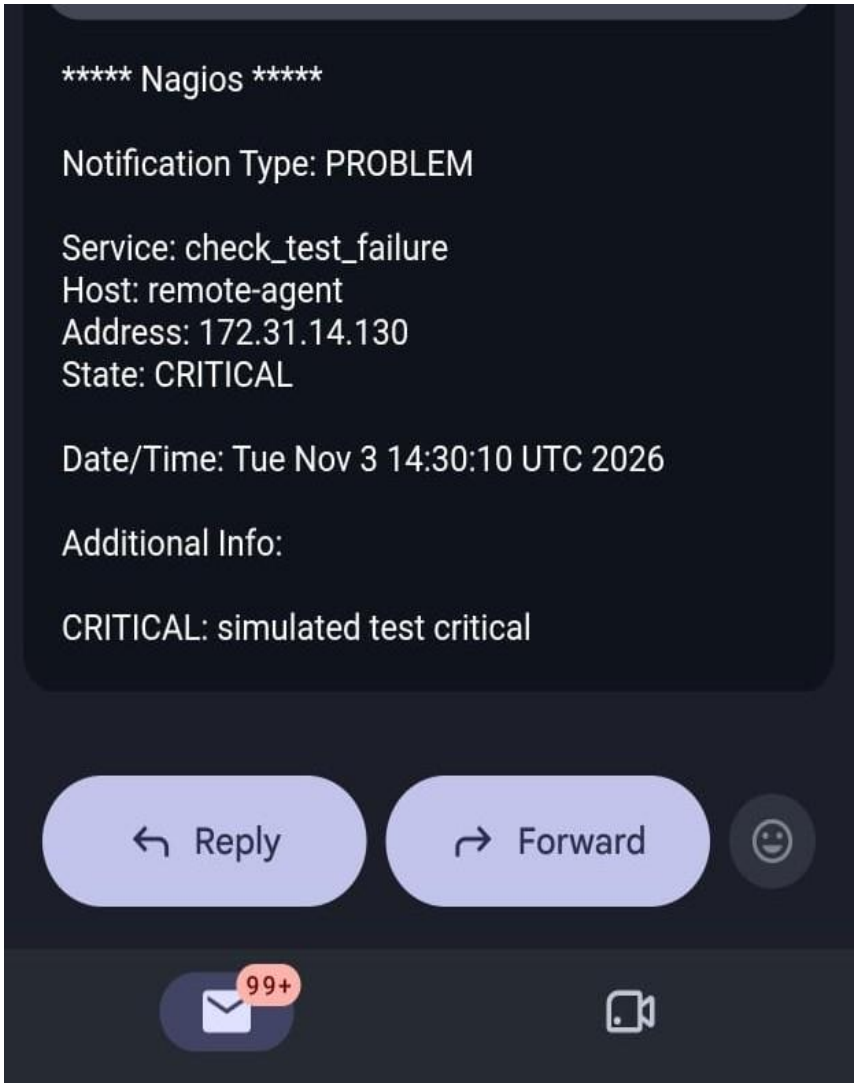


Let’s simulate a failure and recovery:



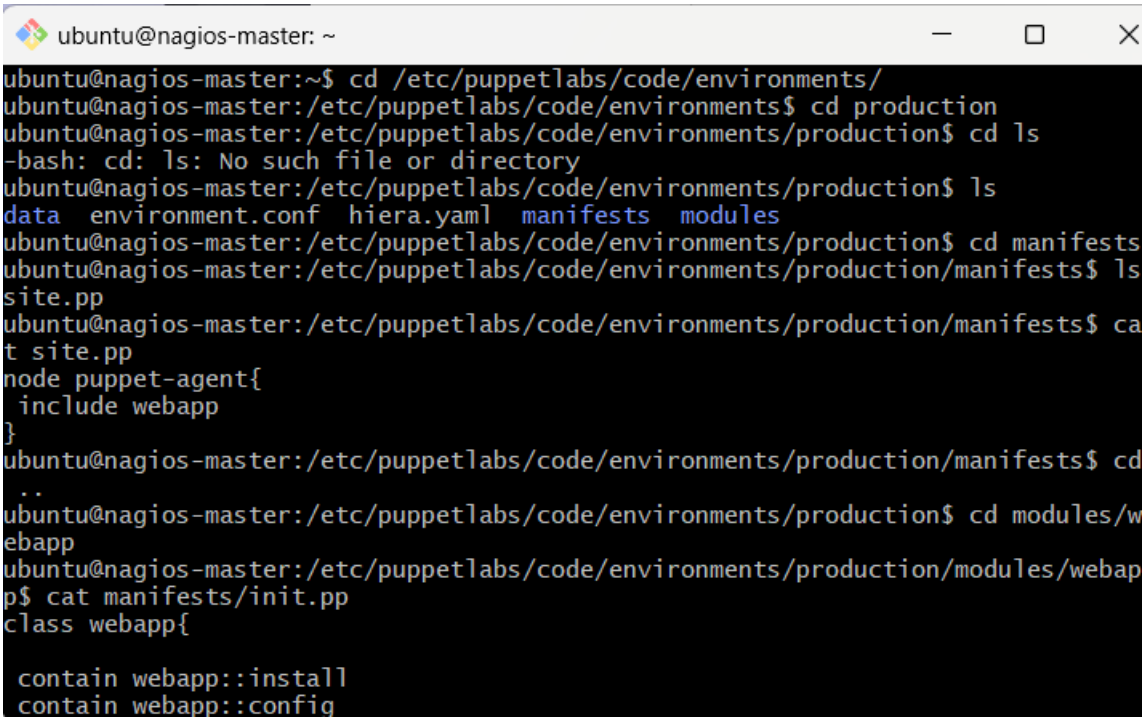

ubuntu@nagios-master: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/agent1.cfg
service_description check_disk_root
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_disk_root
contacts nagiosadmin
}
define service {
use generic-service
host_name agent1
service_description check_test_failure
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_test_failure
contacts nagiosadmin
}
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

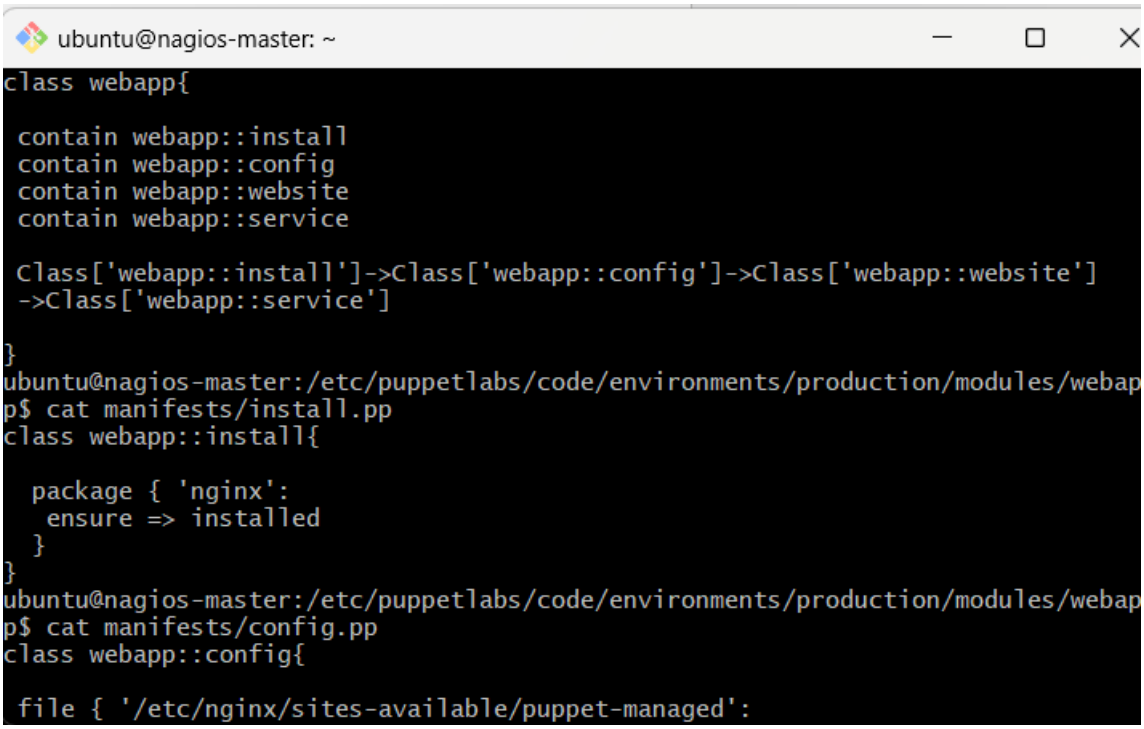

Nagios: 13.203.76.33
Not secure 13.203.76.33/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Comments
Downtime
Service Information
Last Updated: Tue Nov 3 14:34:26 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin
View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service
Service
check_test_failure
On Host
remote-agent
(agent1)
Member of
No servicegroups.
172.31.14.130
Service State Information
Current Status: CRITICAL (for 0d 0h 4m 16s)
Status Information: CRITICAL: simulated test critical
Performance Data:
Current Attempt: 3/3 (HARD state)
Last Check Time: 11-03-2026 14:34:10
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 0.065 seconds
Next Scheduled Check: 11-03-2026 14:35:10
Last State Change: 11-03-2026 14:30:10
Last Notification: 11-03-2026 14:30:10 (notification 1)
Is This Service Flapping? NO (5.72% state change)
In Scheduled Downtime? NO
Last Update: 11-03-2026 14:34:20 (0d 0h 0m 6s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED
Service Commands
Disable active checks of this service
Re-schedule the next check of this service
Submit passive check result for this service
Stop accepting passive checks for this service
Stop obsessing over this service
Acknowledge this service problem
Disable notifications for this service
Delay next service notification
Send custom service notification
Schedule downtime for this service
Disable event handler for this service
Disable flap detection for this service
Clear flapping state for this service
Service Comments
Add a new comment
Delete all comments
Entry Time Author Comment Comment ID Persistent Type Expires Actions
This service has no comments associated with it
Page Tour



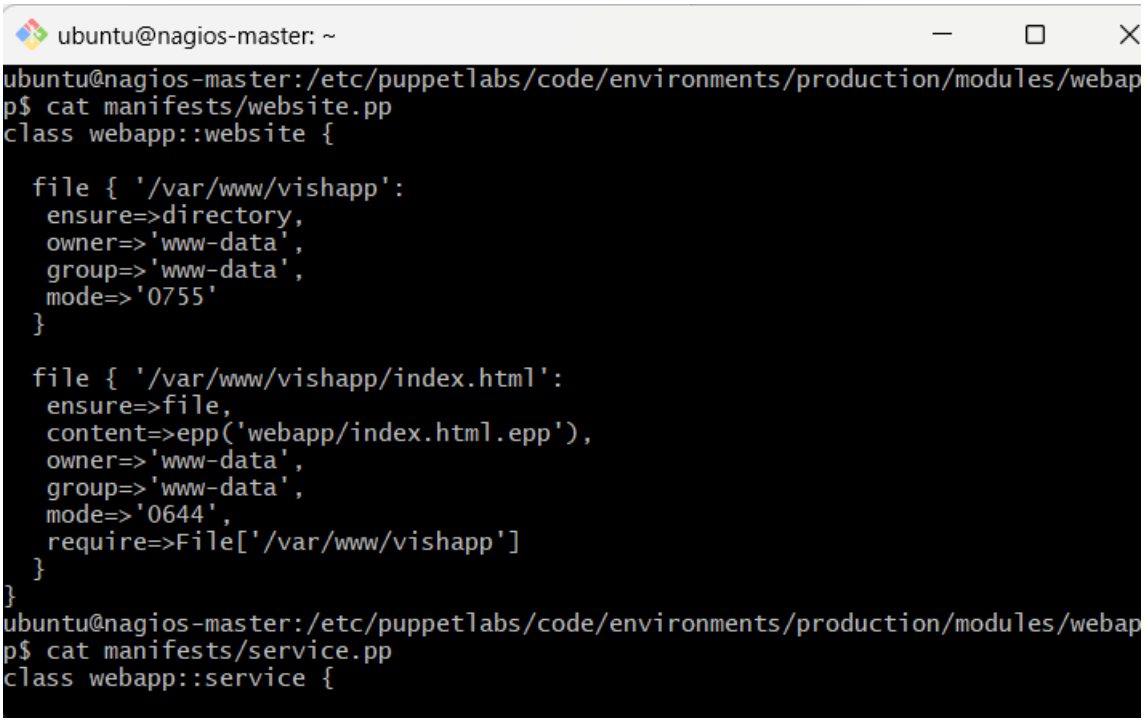
- **From next page (The most interesting Puppet-Nagios integration is documented)**

Enterprise-Grade Puppet and Nagios Integration: End-to-End Infrastructure Automation with Puppet Manifests, Modules, Templates, Package and Service Management, NRPE Configuration, Custom Python HTTP Monitoring Plugins, Failure Simulation, Desired-State Enforcement, Automated Recovery, and Email Notifications

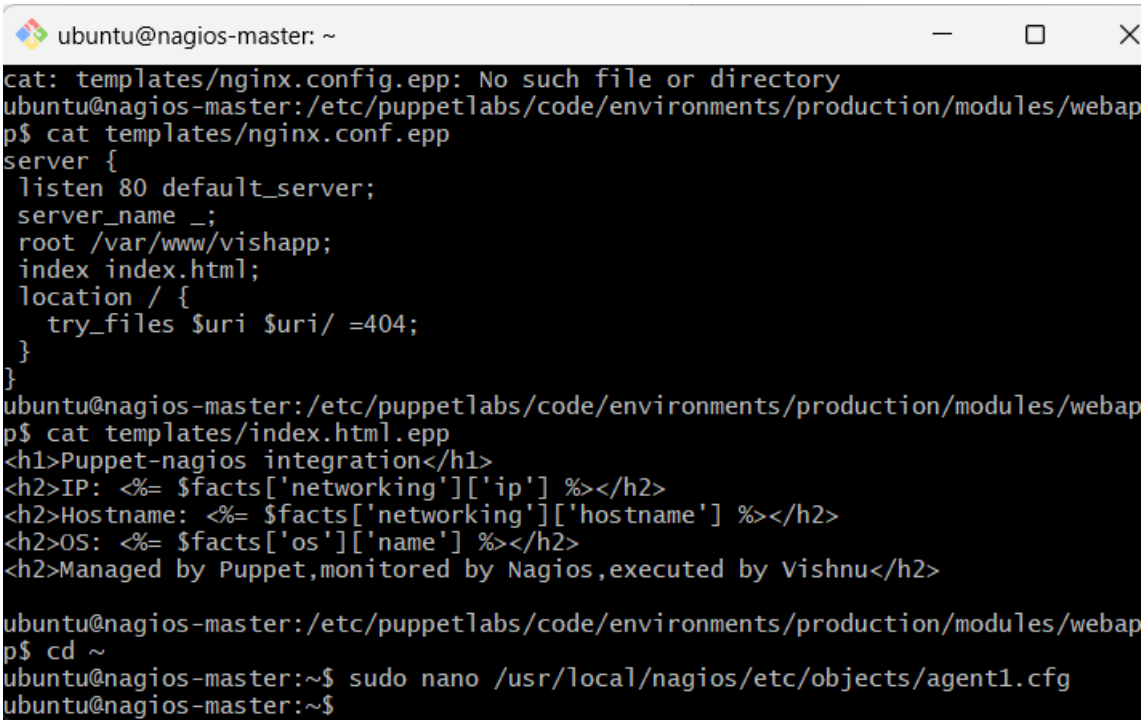


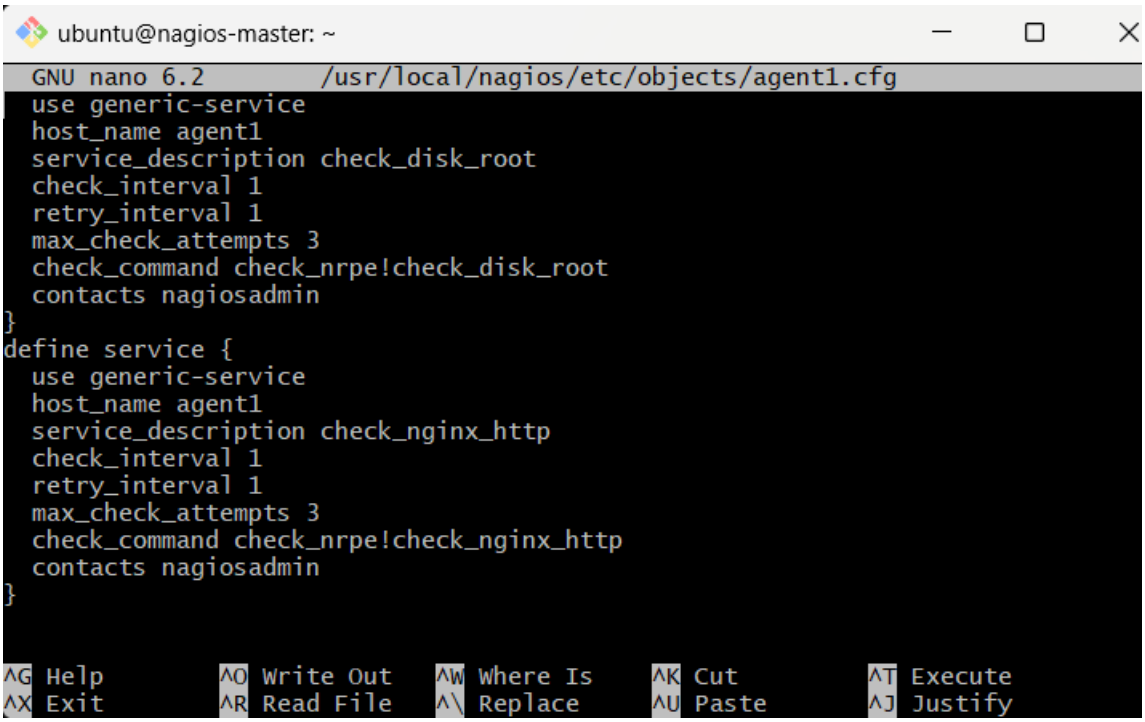

ubuntu@nagios-master: ~
class webapp{
contain webapp::install
contain webapp::config
contain webapp::website
contain webapp::service
Class['webapp::install']->Class['webapp::config']->Class['webapp::website']
->Class['webapp::service']
}
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat manifests/install.pp
class webapp::install{
package { 'nginx':
ensure => installed
}
}
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat manifests/config.pp
class webapp::config{
file { '/etc/nginx/sites-available/puppet-managed':

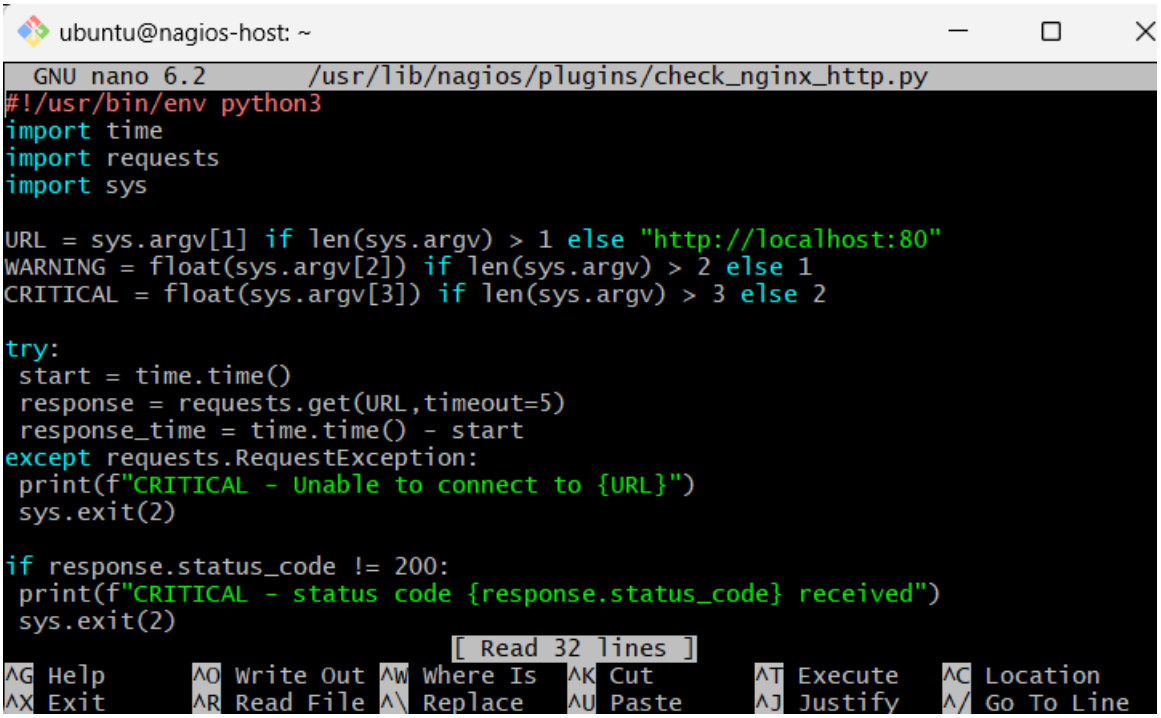

ubuntu@nagios-master: ~
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat manifests/config.pp
class webapp::config{
file { '/etc/nginx/sites-available/puppet-managed':
ensure=>file,
content=>epp('webapp/nginx.conf.epp'),
require=>Package['nginx'],
notify=>Service['nginx']
}
file { '/etc/nginx/sites-enabled/puppet-managed':
ensure=>link,
target=>'/etc/nginx/sites-available/puppet-managed',
require=>File['/etc/nginx/sites-available/puppet-managed'],
notify=>Service['nginx']
}
file { '/etc/nginx/sites-enabled/default':
ensure=>absent,
notify=>Service['nginx']
}
}
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap

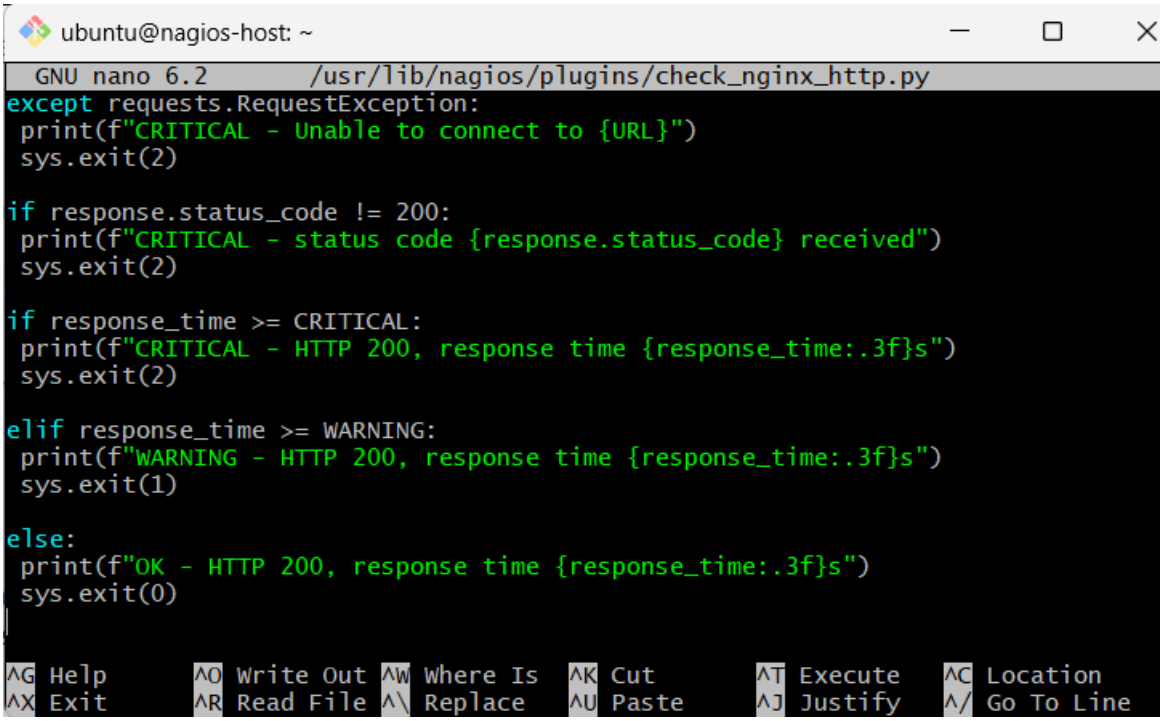

ubuntu@nagios-master: ~
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat manifests/website.pp
class webapp::website {
file { '/var/www/vishapp':
ensure=>directory,
owner=>'www-data',
group=>'www-data',
mode=>'0755'
}
file { '/var/www/vishapp/index.html':
ensure=>file,
content=>epp('webapp/index.html.epp'),
owner=>'www-data',
group=>'www-data',
mode=>'0644',
require=>File['/var/www/vishapp']
}
}
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat manifests/service.pp
class webapp::service {

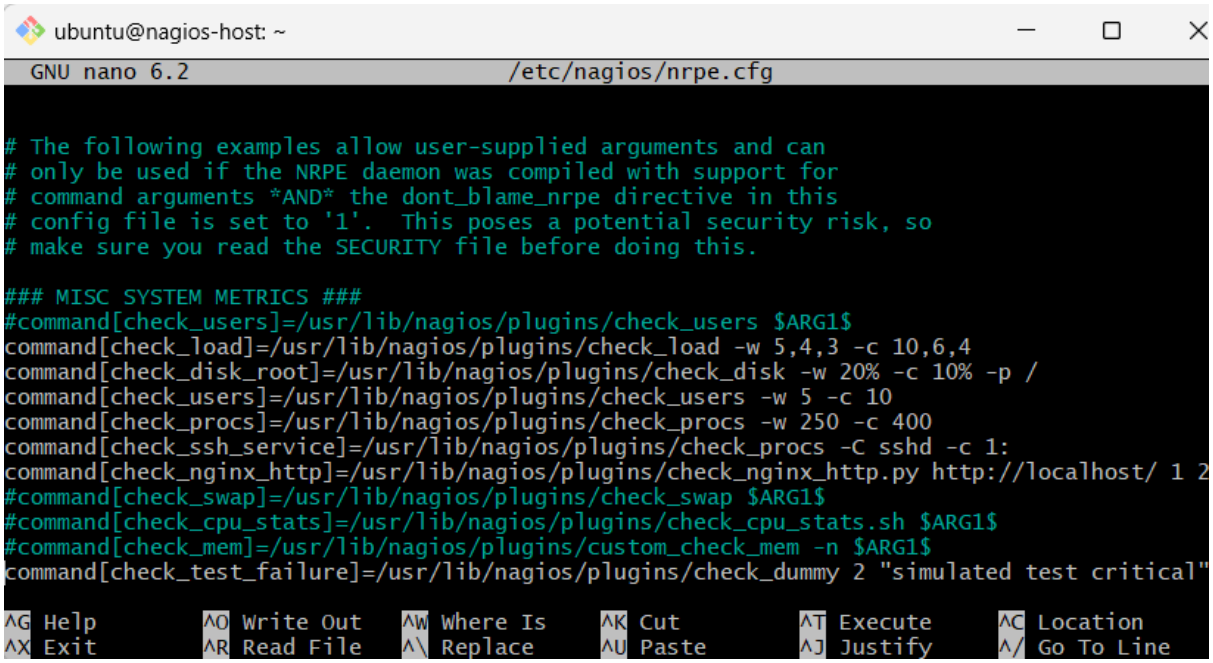

ubuntu@nagios-master: ~
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat manifests/service.pp
class webapp::service {
service { 'nginx':
ensure=>running,
enable=>true,
require=>Package['nginx']
}
}
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat templates/nginx.config.epp
cat: templates/nginx.config.epp: No such file or directory
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat templates/nginx.conf.epp
server {
listen 80 default_server;
server_name _;
root /var/www/vishapp;
index index.html;
location / {
try_files $uri $uri/ =404;
}
}

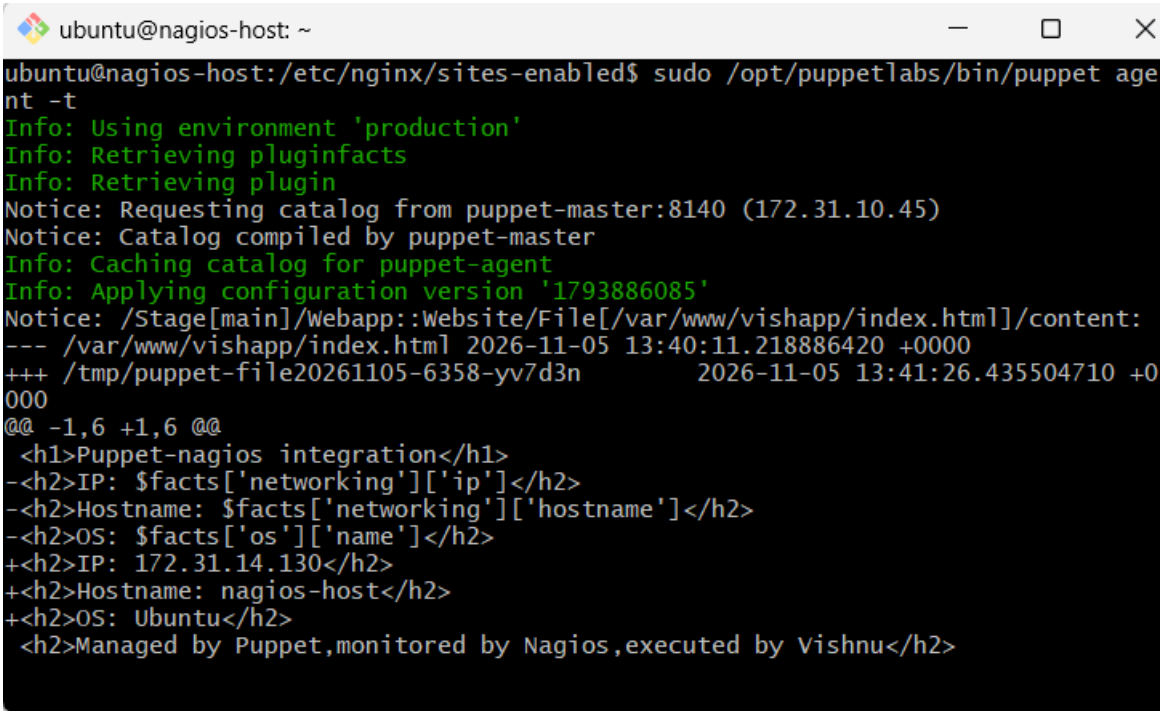

ubuntu@nagios-master: ~
cat: templates/nginx.config.epp: No such file or directory
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat templates/nginx.conf.epp
server {
 listen 80 default_server;
 server_name _;
 root /var/www/vishapp;
 index index.html;
 location / {
   try_files $uri $uri/ =404;
 }
}
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cat templates/index.html.epp
<h1>Puppet-nagios integration</h1>
<h2>IP: <%= $facts['networking']['ip'] %></h2>
<h2>Hostname: <%= $facts['networking']['hostname'] %></h2>
<h2>OS: <%= $facts['os']['name'] %></h2>
<h2>Managed by Puppet,monitored by Nagios,executed by Vishnu</h2>
ubuntu@nagios-master:/etc/puppetlabs/code/environments/production/modules/webap
p$ cd ~
ubuntu@nagios-master:~$ sudo nano /usr/local/nagios/etc/objects/agent1.cfg
ubuntu@nagios-master:~$

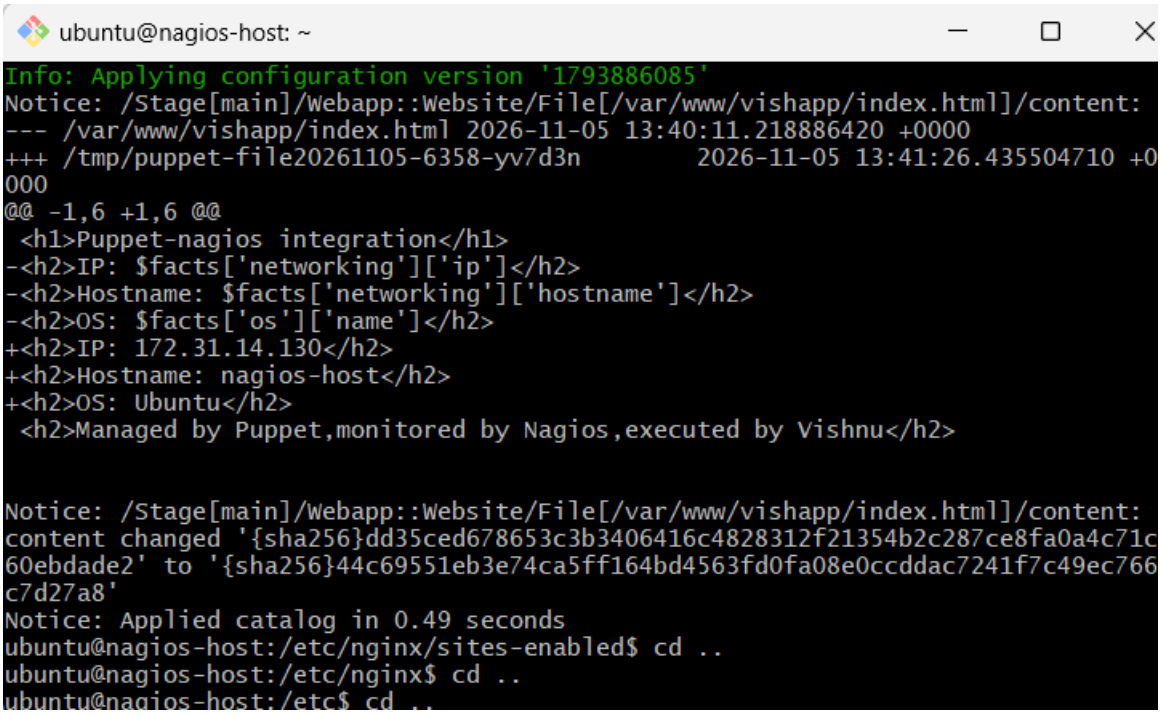

ubuntu@nagios-master: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/agent1.cfg
use generic-service
host_name agent1
service_description check_disk_root
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_disk_root
contacts nagiosadmin
}
define service {
use generic-service
host_name agent1
service_description check_nginx_http
check_interval 1
retry_interval 1
max_check_attempts 3
check_command check_nrpe!check_nginx_http
contacts nagiosadmin
}
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

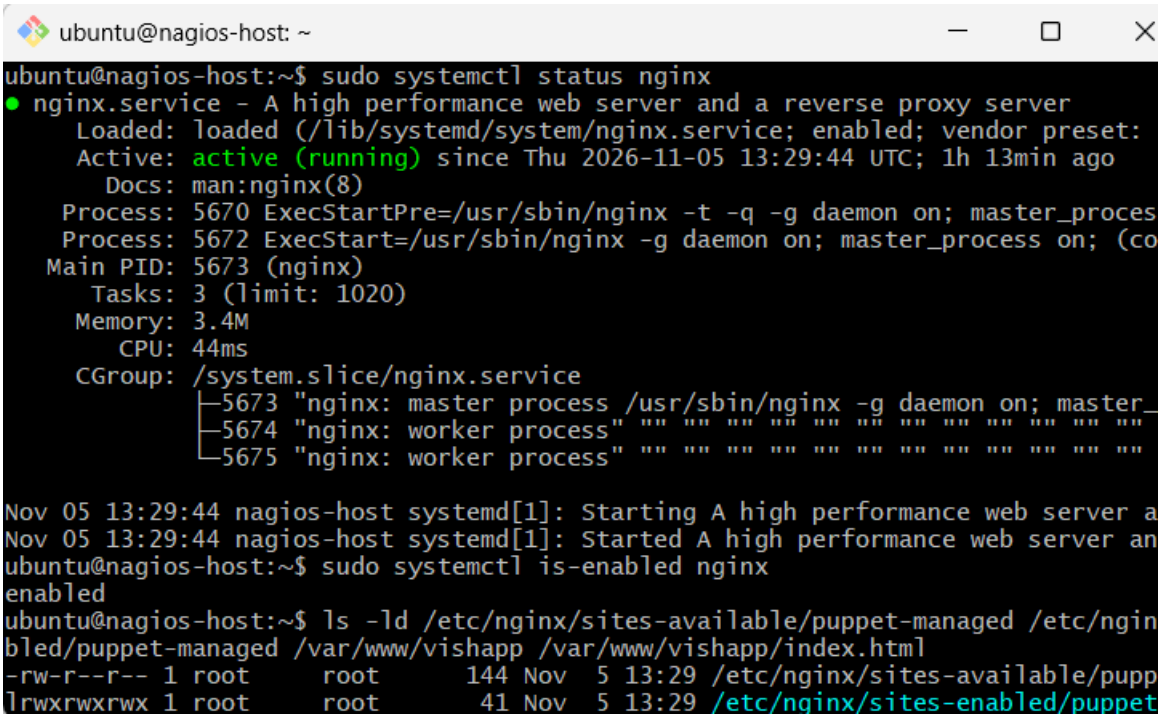

ubuntu@nagios-host: ~
GNU nano 6.2 /usr/lib/nagios/plugins/check_nginx_http.py
#!/usr/bin/env python3
import time
import requests
import sys
URL = sys.argv[1] if len(sys.argv) > 1 else "http://localhost:80"
WARNING = float(sys.argv[2]) if len(sys.argv) > 2 else 1
CRITICAL = float(sys.argv[3]) if len(sys.argv) > 3 else 2
try:
start = time.time()
response = requests.get(URL,timeout=5)
response_time = time.time() - start
except requests.RequestException:
print(f"CRITICAL - Unable to connect to {URL}")
sys.exit(2)
if response.status_code != 200:
print(f"CRITICAL - status code {response.status_code} received")
sys.exit(2)
[Read 32 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

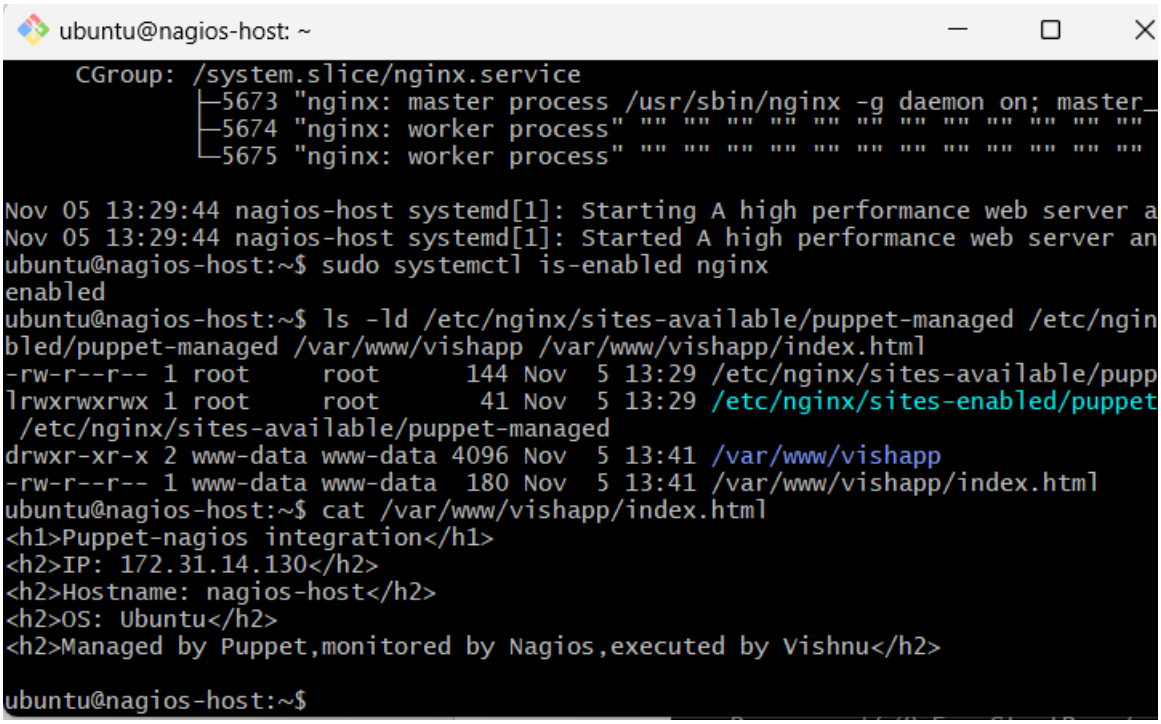

ubuntu@nagios-host: ~
GNU nano 6.2 /usr/lib/nagios/plugins/check_nginx_http.py
except requests.RequestException:
print(f"CRITICAL - Unable to connect to {URL}")
sys.exit(2)
if response.status_code != 200:
print(f"CRITICAL - status code {response.status_code} received")
sys.exit(2)
if response_time >= CRITICAL:
print(f"CRITICAL - HTTP 200, response time {response_time:.3f}s")
sys.exit(2)
elif response_time >= WARNING:
print(f"WARNING - HTTP 200, response time {response_time:.3f}s")
sys.exit(1)
else:
print(f"OK - HTTP 200, response time {response_time:.3f}s")
sys.exit(0)
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

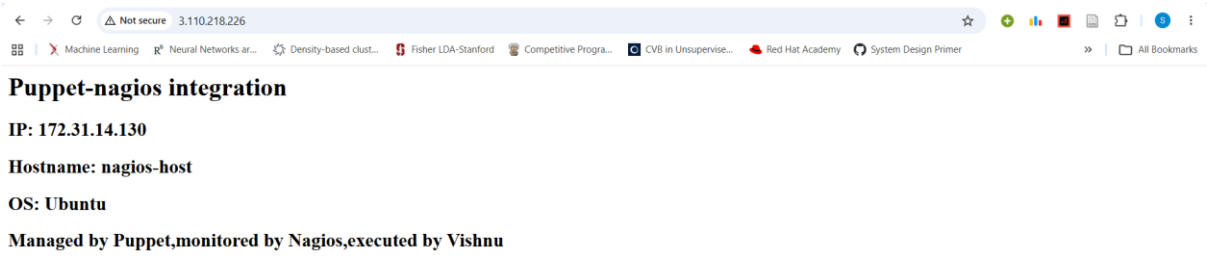

ubuntu@nagios-host: ~
GNU nano 6.2 /etc/nagios/nrpe.cfg
The following examples allow user-supplied arguments and can
only be used if the NRPE daemon was compiled with support for
command arguments *AND* the dont_blame_nrpe directive in this
config file is set to '1'. This poses a potential security risk, so
make sure you read the SECURITY file before doing this.
MISC SYSTEM METRICS
#command[check_users]=/usr/lib/nagios/plugins/check_users $ARG1$
command[check_load]=/usr/lib/nagios/plugins/check_load -w 5,4,3 -c 10,6,4
command[check_disk_root]=/usr/lib/nagios/plugins/check_disk -w 20% -c 10% -p /
command[check_users]=/usr/lib/nagios/plugins/check_users -w 5 -c 10
command[check_procs]=/usr/lib/nagios/plugins/check_procs -w 250 -c 400
command[check_ssh_service]=/usr/lib/nagios/plugins/check_procs -C sshd -c 1:
command[check_nginx_http]=/usr/lib/nagios/plugins/check_nginx_http.py http://localhost/ 1 2
#command[check_swap]=/usr/lib/nagios/plugins/check_swap $ARG1$
#command[check_cpu_stats]=/usr/lib/nagios/plugins/check_cpu_stats.sh $ARG1$
#command[check_mem]=/usr/lib/nagios/plugins/custom_check_mem -n $ARG1$
command[check_test_failure]=/usr/lib/nagios/plugins/check_dummy 2 "simulated test critical"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

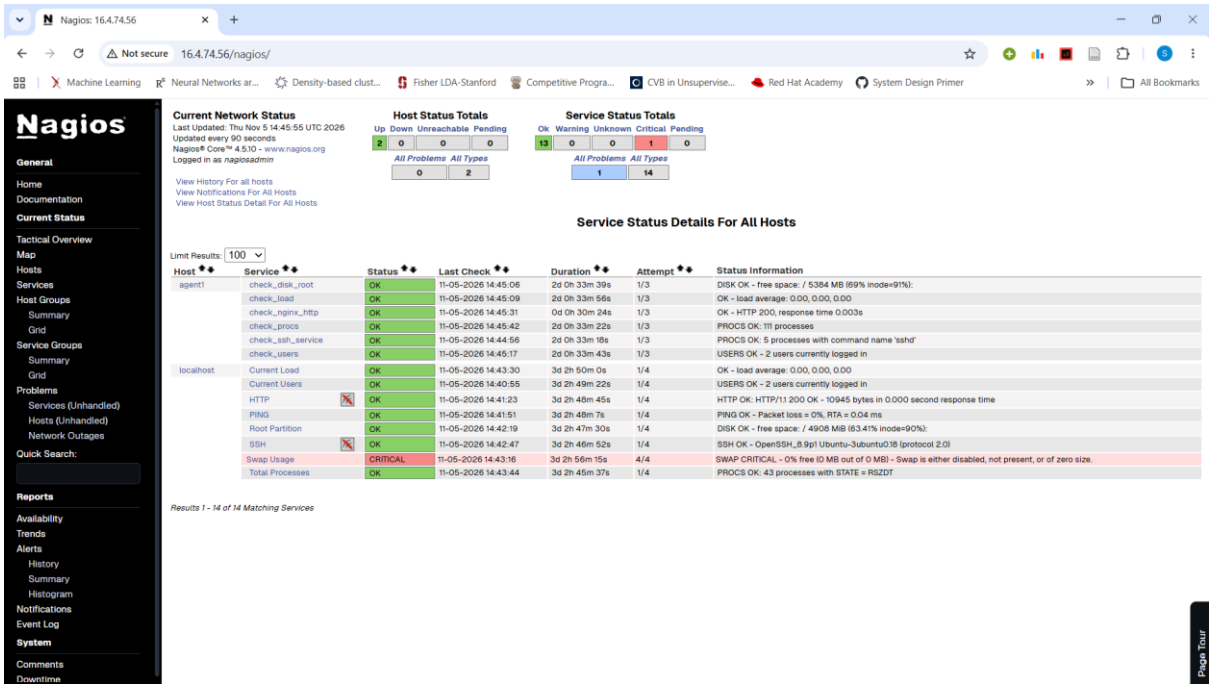

ubuntu@nagios-host: ~
ubuntu@nagios-host:/etc/nginx/sites-enabled$ sudo /opt/puppetlabs/bin/puppet agent -t
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Notice: Requesting catalog from puppet-master:8140 (172.31.10.45)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent
Info: Applying configuration version '1793886085'
Notice: /Stage[main]/Webapp::Website/File[/var/www/vishapp/index.html]/content:
--- /var/www/vishapp/index.html 2026-11-05 13:40:11.218886420 +0000
+++ /tmp/puppet-file20261105-6358-yv7d3n 2026-11-05 13:41:26.435504710 +0000
@@ -1,6 +1,6 @@
<h1>Puppet-nagios integration</h1>
-<h2>IP: $facts['networking']['ip']</h2>
-<h2>Hostname: $facts['networking']['hostname']</h2>
-<h2>OS: $facts['os']['name']</h2>
+<h2>IP: 172.31.14.130</h2>
+<h2>Hostname: nagios-host</h2>
+<h2>OS: Ubuntu</h2>
<h2>Managed by Puppet,monitored by Nagios,executed by Vishnu</h2>

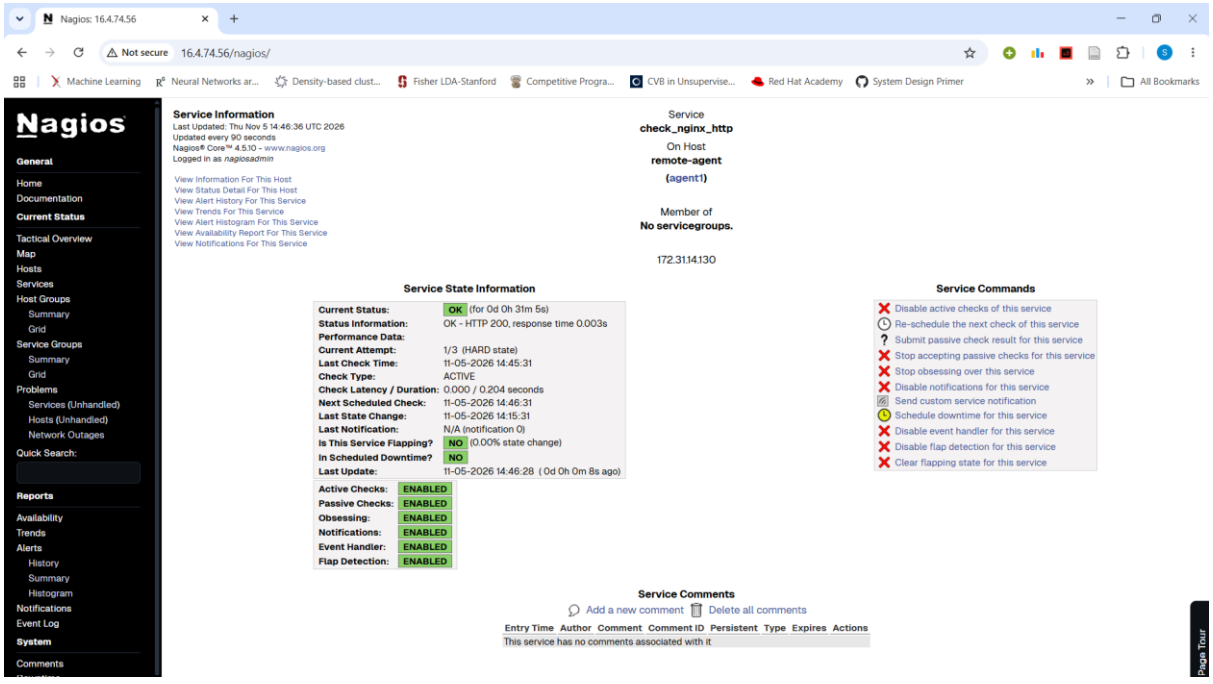

ubuntu@nagios-host: ~
Info: Applying configuration version '1793886085'
Notice: /Stage[main]/Webapp::Website/File[/var/www/vishapp/index.html]/content:
--- /var/www/vishapp/index.html 2026-11-05 13:40:11.218886420 +0000
+++ /tmp/puppet-file20261105-6358-yv7d3n 2026-11-05 13:41:26.435504710 +0
000
@@ -1,6 +1,6 @@
<h1>Puppet-nagios integration</h1>
-<h2>IP: $facts['networking']['ip']</h2>
-<h2>Hostname: $facts['networking']['hostname']</h2>
-<h2>OS: $facts['os']['name']</h2>
+<h2>IP: 172.31.14.130</h2>
+<h2>Hostname: nagios-host</h2>
+<h2>OS: Ubuntu</h2>
<h2>Managed by Puppet,monitored by Nagios,executed by Vishnu</h2>
Notice: /Stage[main]/Webapp::Website/File[/var/www/vishapp/index.html]/content:
content changed '{sha256}dd35ced678653c3b3406416c4828312f21354b2c287ce8fa0a4c71c
60ebdade2' to '{sha256}44c69551eb3e74ca5ff164bd4563fd0fa08e0ccddac7241f7c49ec766
c7d27a8'
Notice: Applied catalog in 0.49 seconds
ubuntu@nagios-host:/etc/nginx/sites-enabled$ cd ..
ubuntu@nagios-host:/etc/nginx$ cd ..
ubuntu@nagios-host:/etc$ cd ..


ubuntu@nagios-host: ~
ubuntu@nagios-host:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset:
Active: active (running) since Thu 2026-11-05 13:29:44 UTC; 1h 13min ago
Docs: man:nginx(8)
Process: 5670 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_proces
Process: 5672 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (co
Main PID: 5673 (nginx)
Tasks: 3 (limit: 1020)
Memory: 3.4M
CPU: 44ms
CGroup: /system.slice/nginx.service
├─5673 "nginx: master process /usr/sbin/nginx -g daemon on; master_
├─5674 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" ""
└─5675 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" ""
Nov 05 13:29:44 nagios-host systemd[1]: Starting A high performance web server a
Nov 05 13:29:44 nagios-host systemd[1]: Started A high performance web server an
ubuntu@nagios-host:~$ sudo systemctl is-enabled nginx
enabled
ubuntu@nagios-host:~$ ls -ld /etc/nginx/sites-available/puppet-managed /etc/ngin
bled/puppet-managed /var/www/vishapp /var/www/vishapp/index.html
-rw-r--r-- 1 root root 144 Nov 5 13:29 /etc/nginx/sites-available/pupp
lrwxrwxrwx 1 root root 41 Nov 5 13:29 /etc/nginx/sites-enabled/puppet


ubuntu@nagios-host: ~
CGroup: /system.slice/nginx.service
├─5673 "nginx: master process /usr/sbin/nginx -g daemon on; master_
├─5674 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" "" ""
└─5675 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" "" ""
Nov 05 13:29:44 nagios-host systemd[1]: Starting A high performance web server a
Nov 05 13:29:44 nagios-host systemd[1]: Started A high performance web server an
ubuntu@nagios-host:~$ sudo systemctl is-enabled nginx
enabled
ubuntu@nagios-host:~$ ls -ld /etc/nginx/sites-available/puppet-managed /etc/ngin
bled/puppet-managed /var/www/vishapp /var/www/vishapp/index.html
-rw-r--r-- 1 root root 144 Nov 5 13:29 /etc/nginx/sites-available/pupp
lrwxrwxrwx 1 root root 41 Nov 5 13:29 /etc/nginx/sites-enabled/puppet
/etc/nginx/sites-available/puppet-managed
drwxr-xr-x 2 www-data www-data 4096 Nov 5 13:41 /var/www/vishapp
-rw-r--r-- 1 www-data www-data 180 Nov 5 13:41 /var/www/vishapp/index.html
ubuntu@nagios-host:~$ cat /var/www/vishapp/index.html
<h1>Puppet-nagios integration</h1>
<h2>IP: 172.31.14.130</h2>
<h2>Hostname: nagios-host</h2>
<h2>OS: Ubuntu</h2>
<h2>Managed by Puppet,monitored by Nagios,executed by Vishnu</h2>
ubuntu@nagios-host:~$

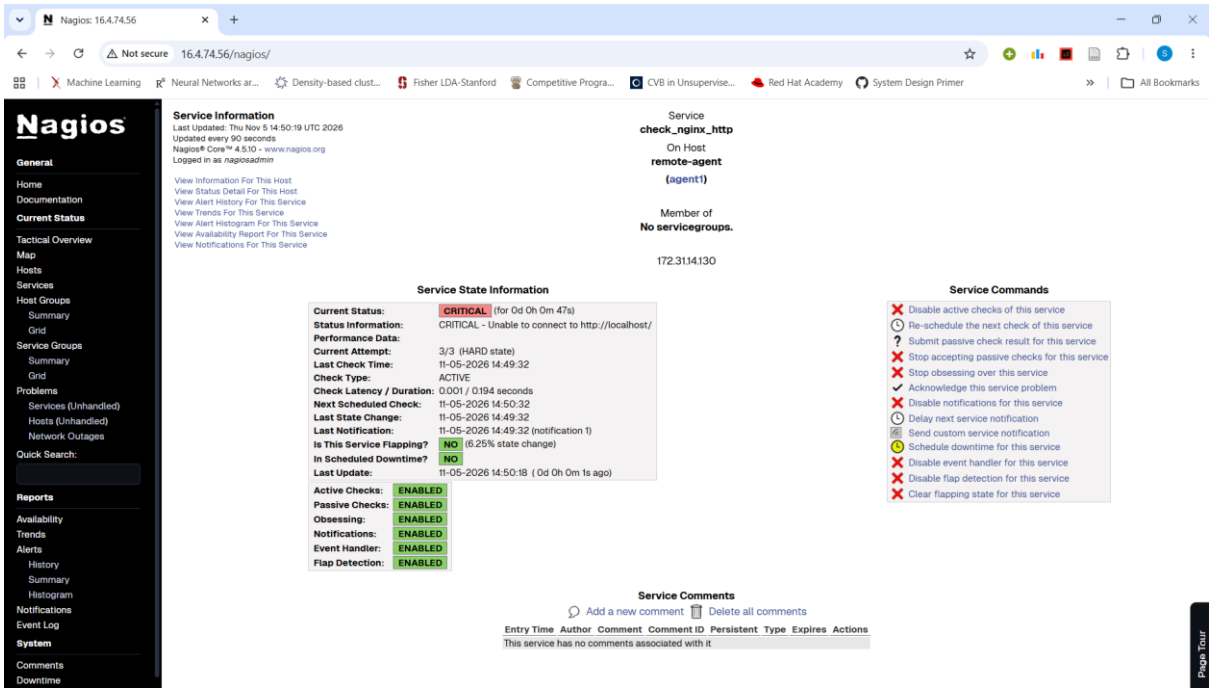

Not secure
3.110.218.226
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmarks
Puppet-nagios integration
IP: 172.31.14.130
Hostname: nagios-host
OS: Ubuntu
Managed by Puppet,monitored by Nagios,executed by Vishnu

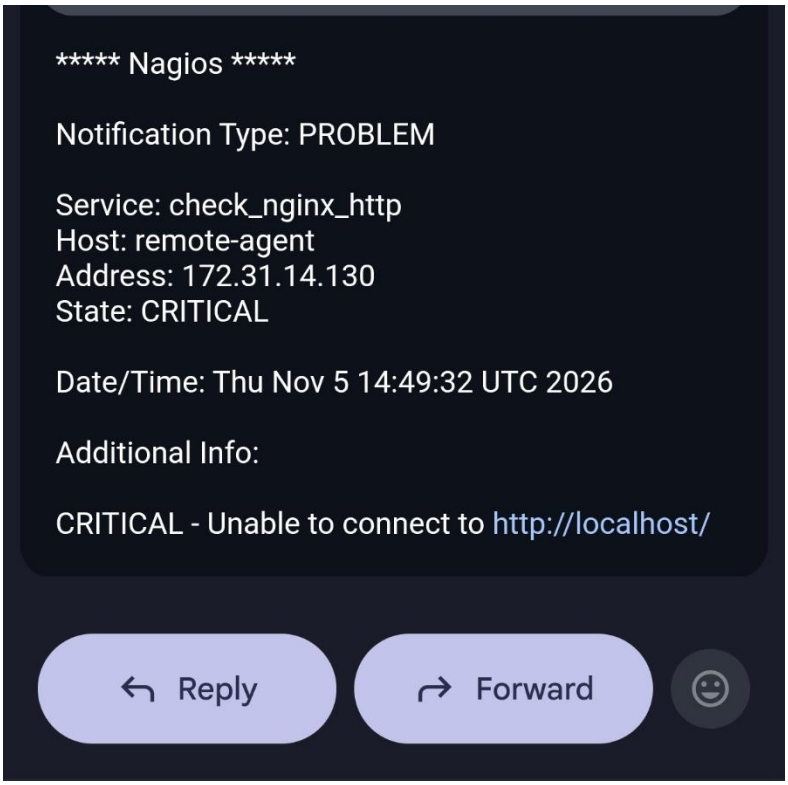


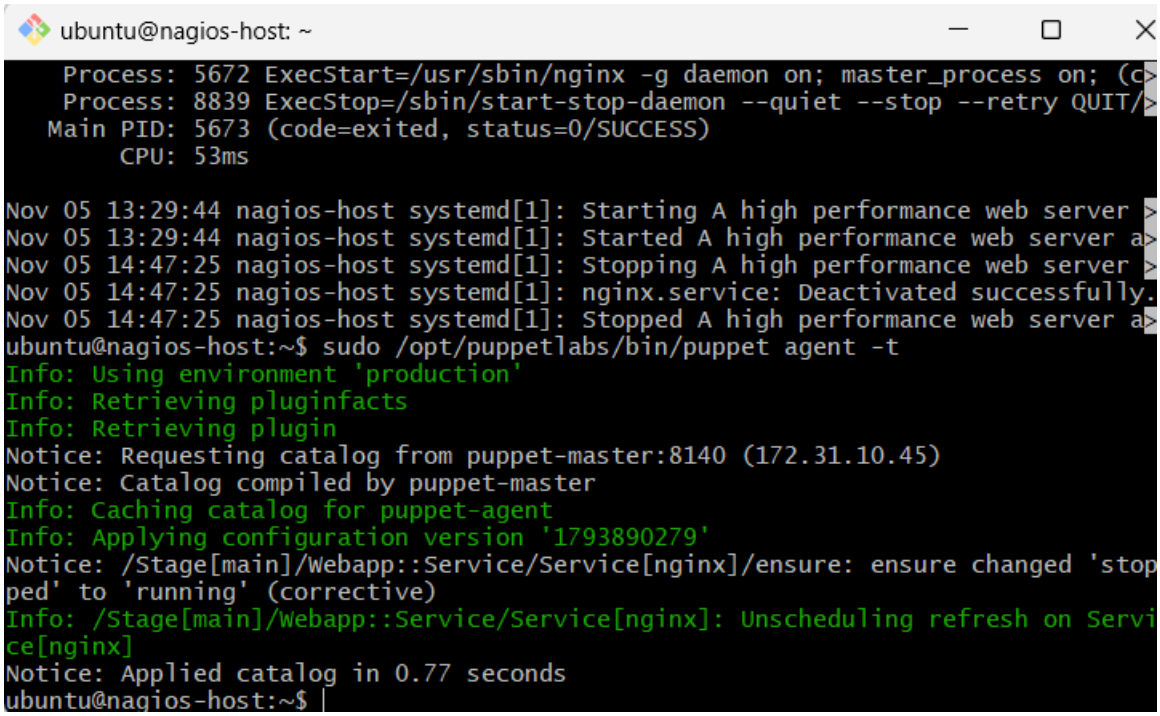


Let's simulate criticality:

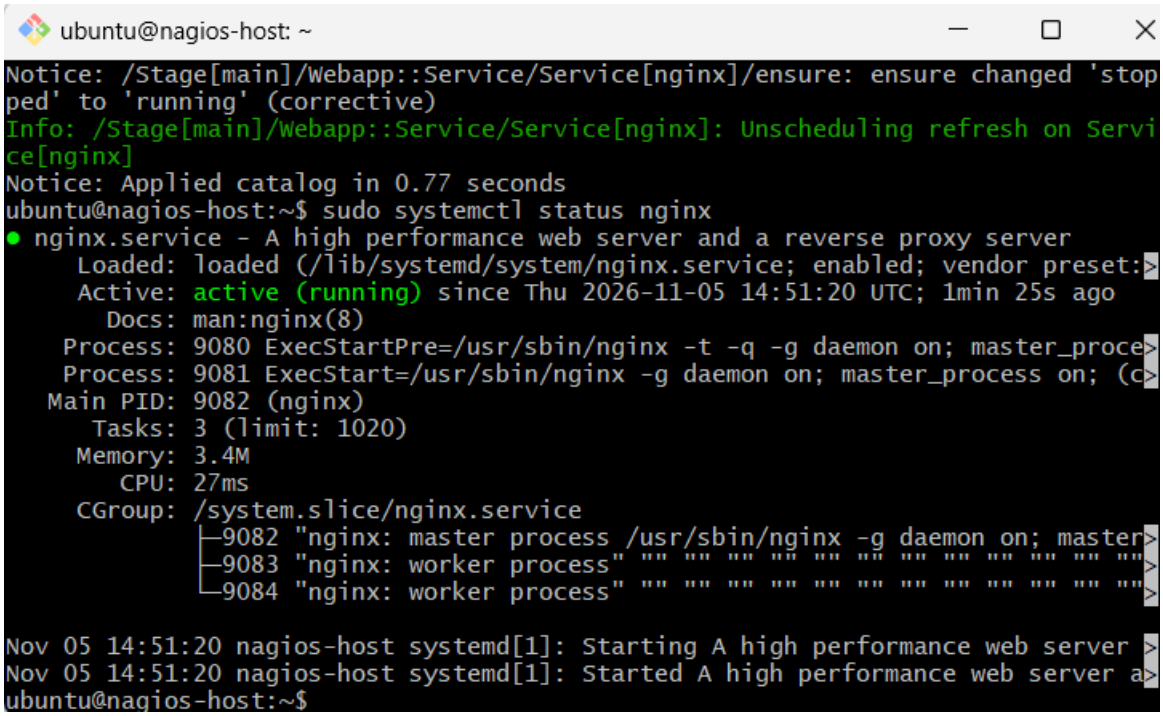
Simulated Failure → Nagios CRITICAL Alert → Puppet Desired-State Enforcement → Nginx Recovery → Nagios Recovery Notification

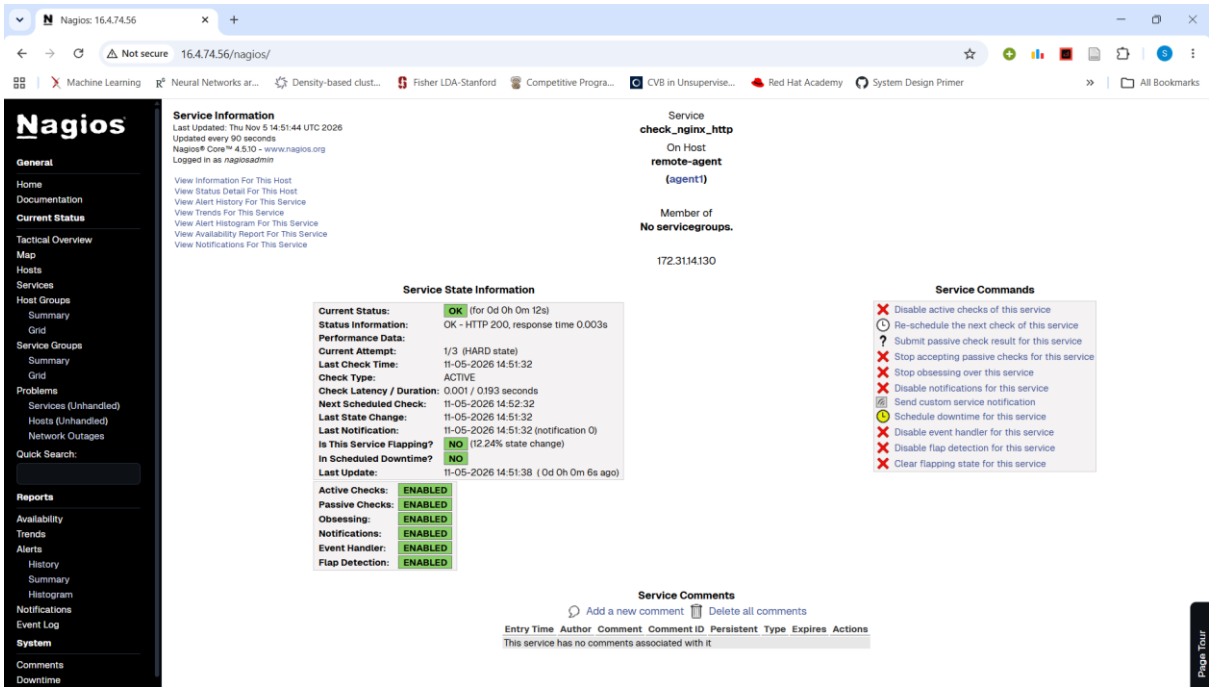

Nagios: 16.4.74.56
Not secure
16.4.74.56/nagios/
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmarks
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Comments
Downtime
Service Information
Last Updated: Thu Nov 5 14:50:19 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin
View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service
Service
check_nginx_http
On Host
remote-agent
(agent1)
Member of
No servicegroups.
172.31.14.130
Service State Information
Current Status: CRITICAL (for 0d 0h 0m 47s)
Status Information: CRITICAL - Unable to connect to http://localhost/
Performance Data:
Current Attempt: 3/3 (HARD state)
Last Check Time: 11-05-2026 14:49:32
Check Type: ACTIVE
Check Latency / Duration: 0.001 / 0.194 seconds
Next Scheduled Check: 11-05-2026 14:50:32
Last State Change: 11-05-2026 14:49:32
Last Notification: 11-05-2026 14:49:32 (notification 1)
Is This Service Flapping? NO (6.25% state change)
In Scheduled Downtime? NO
Last Update: 11-05-2026 14:50:18 (0d 0h 0m 1s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED
Service Commands
Disable active checks of this service
Re-schedule the next check of this service
Submit passive check result for this service
Stop accepting passive checks for this service
Stop obsessing over this service
Acknowledge this service problem
Disable notifications for this service
Delay next service notification
Send custom service notification
Schedule downtime for this service
Disable event handler for this service
Disable flap detection for this service
Clear flapping state for this service
Service Comments
Add a new comment
Delete all comments
Entry Time Author Comment Comment ID Persistent Type Expires Actions
This service has no comments associated with it
Page Tour

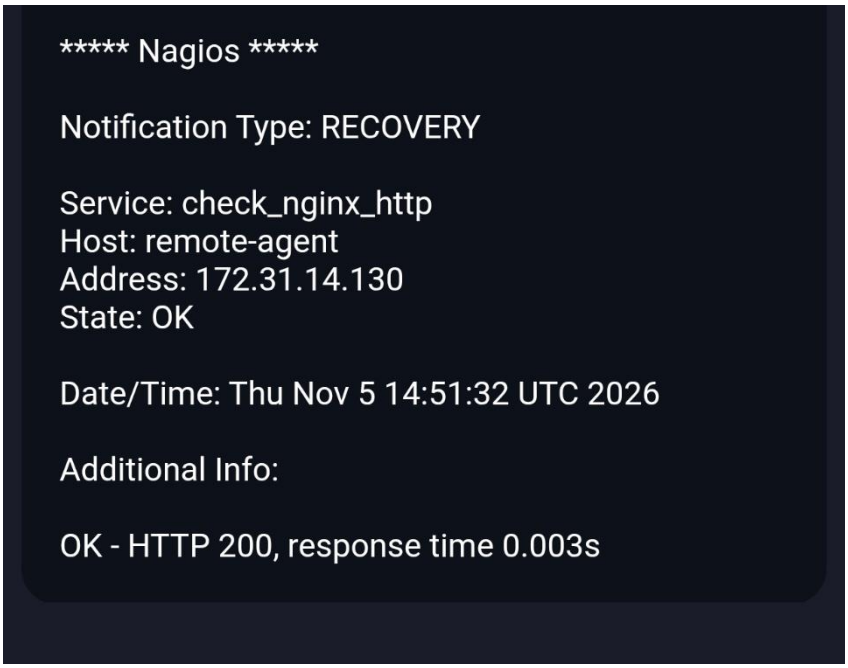

***** Nagios *****
Notification Type: PROBLEM
Service: check_nginx_http
Host: remote-agent
Address: 172.31.14.130
State: CRITICAL
Date/Time: Thu Nov 5 14:49:32 UTC 2026
Additional Info:
CRITICAL - Unable to connect to http://localhost/
Reply
Forward



And there we go, Puppet to enforce the state, there is a reason why DevOps is 'DevOps'







And yea got the recovery email, failure detected by Nagios, state enforced by puppet and recovery triggered successfully.